

Reminders

Upcoming due dates

Wed Assignment 1

Project review (1 per group)

Fri Discussion lab 3

Repo invites! Click accept before it expires!

Section this week covers data visualization

Exploratory Data Analysis (and map making!)

Data Science in Practice

Jason G. Fleischer, PhD

Dept. of Cognitive Science

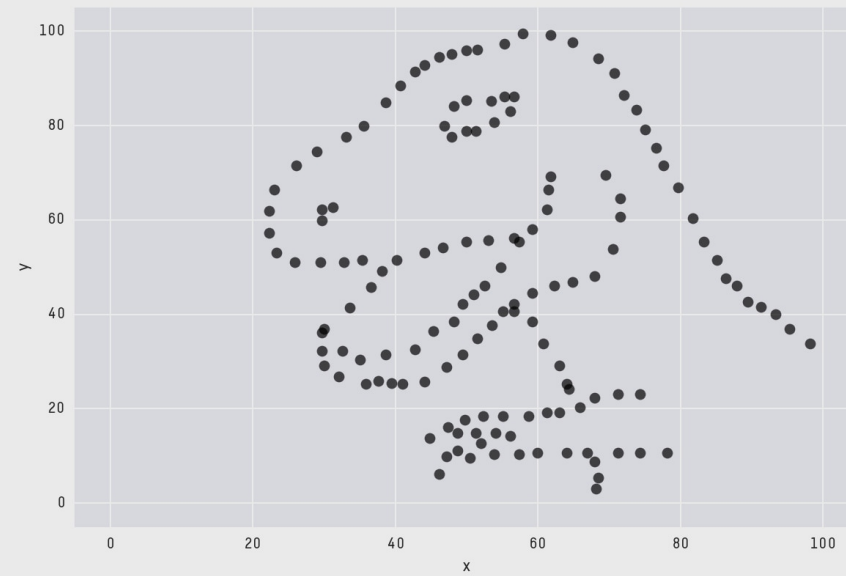
UC San Diego

<https://jgfleischer.com>

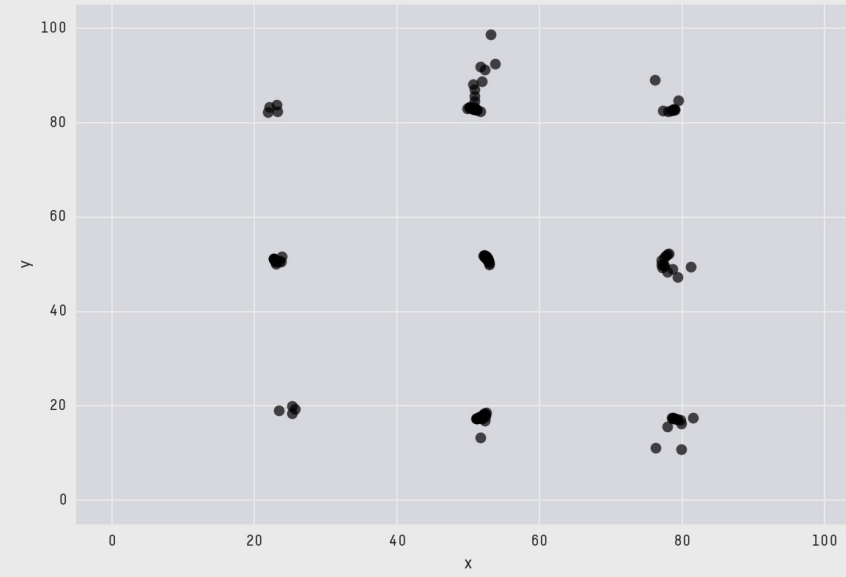
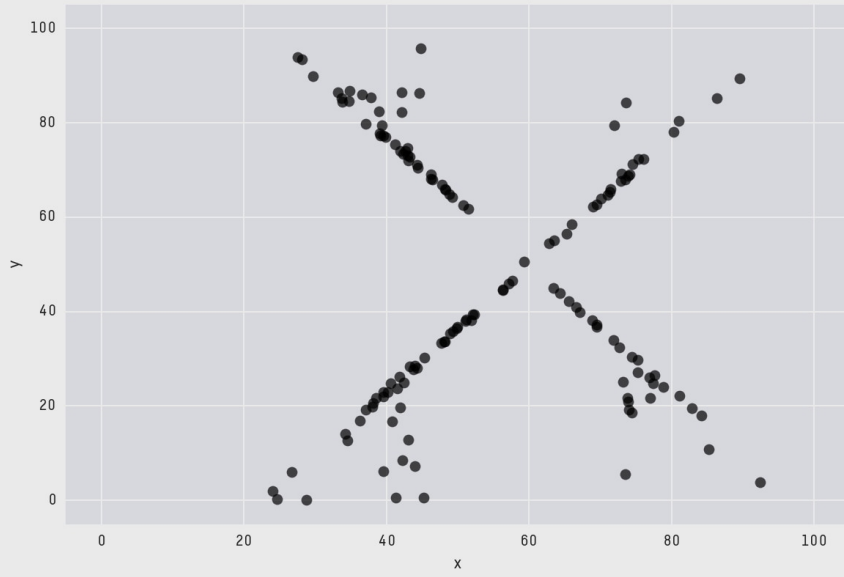
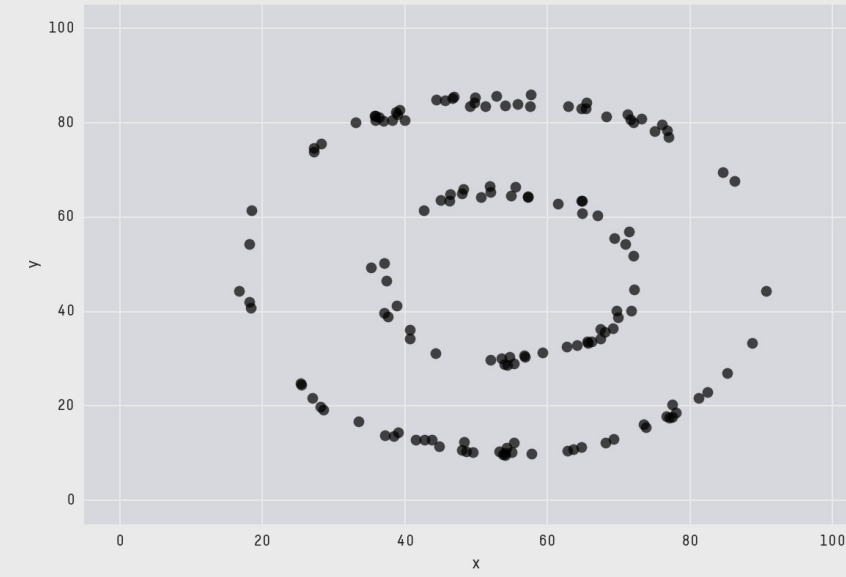
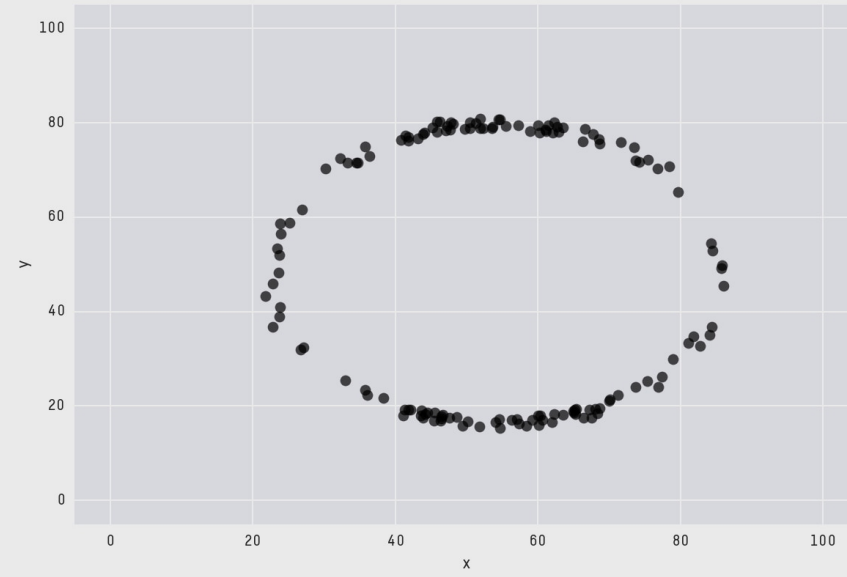
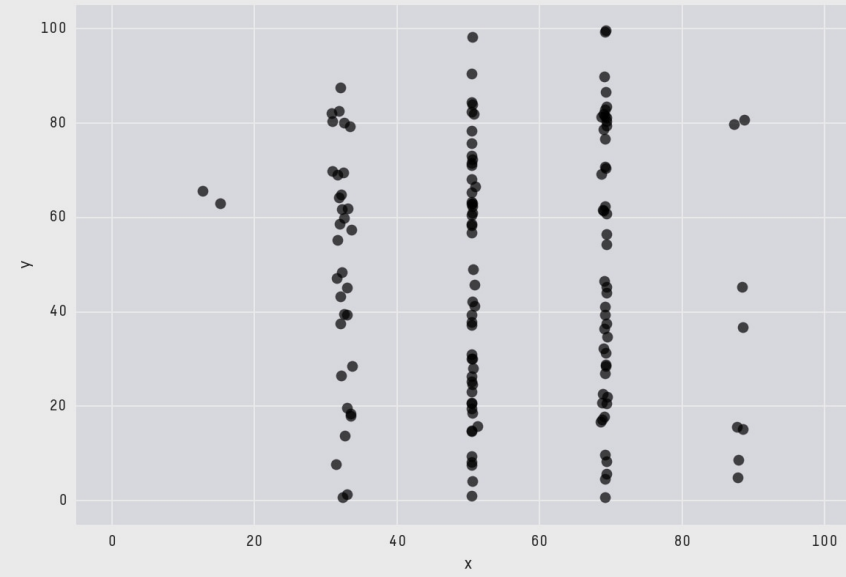
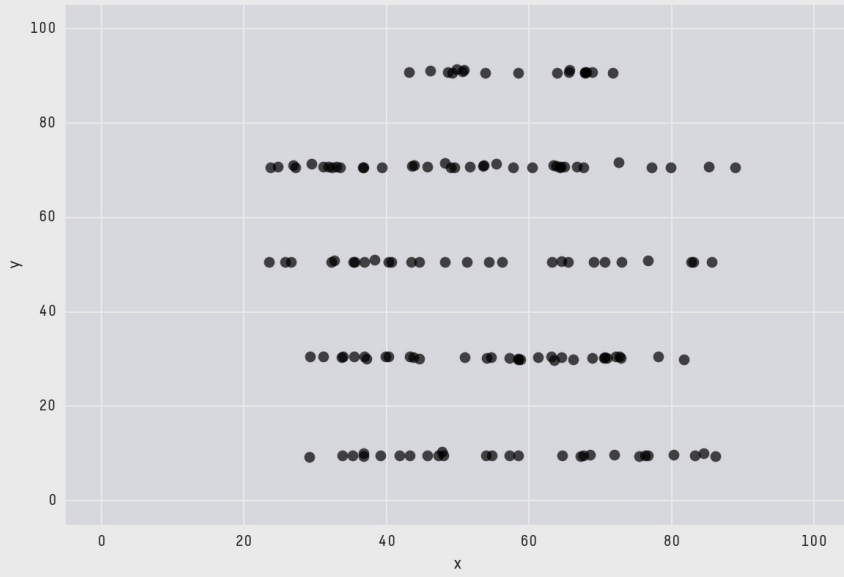
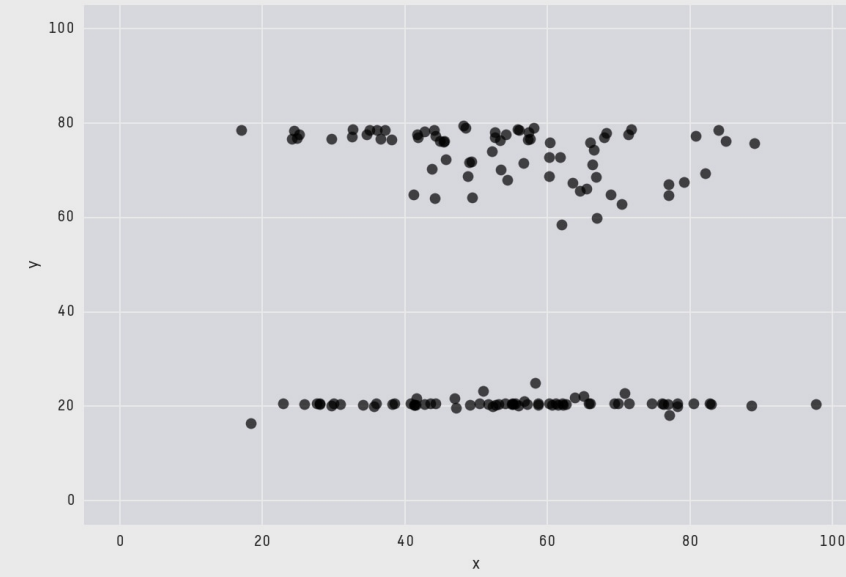
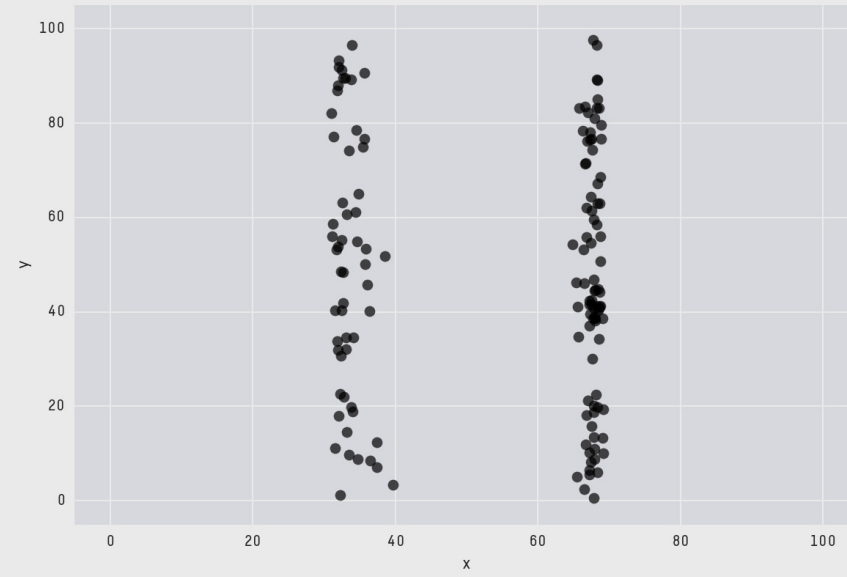
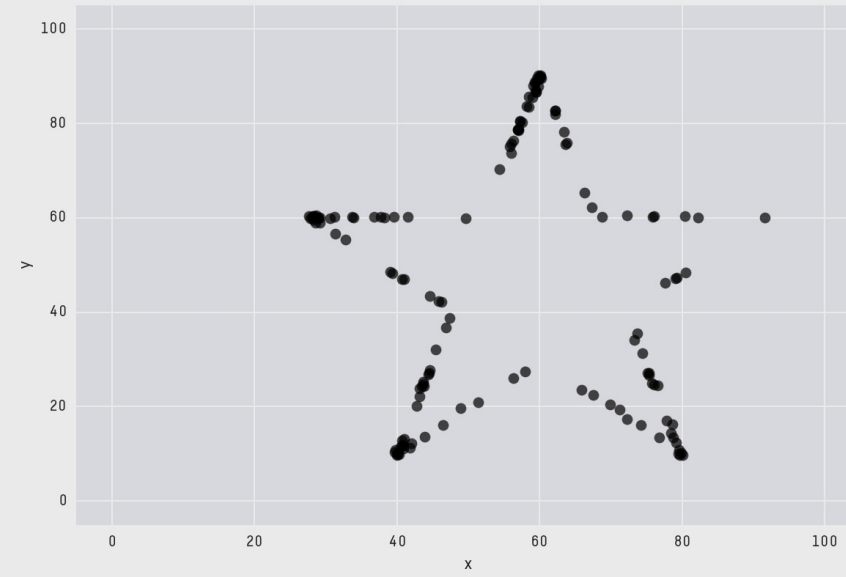
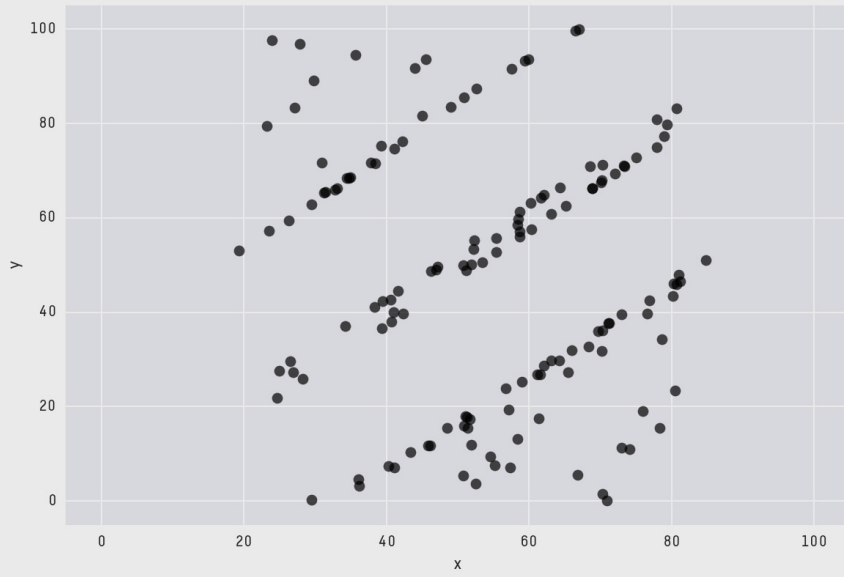
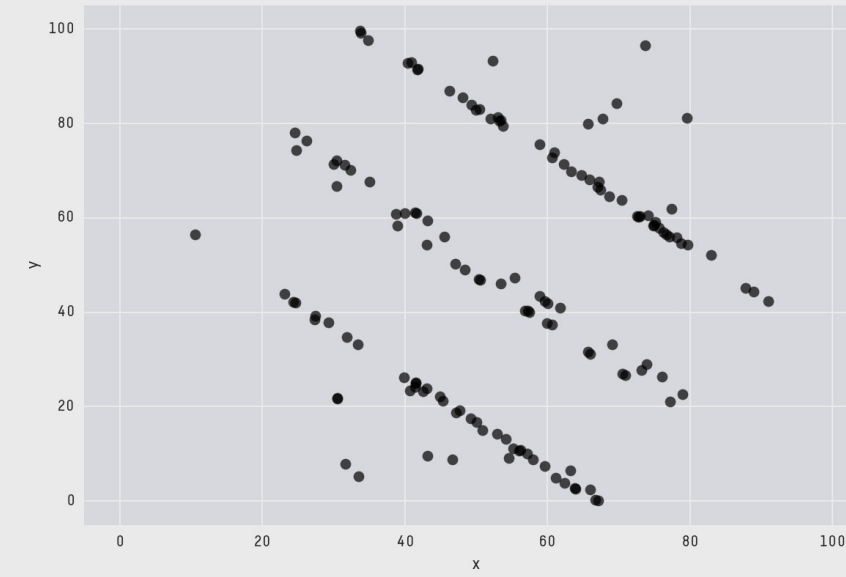
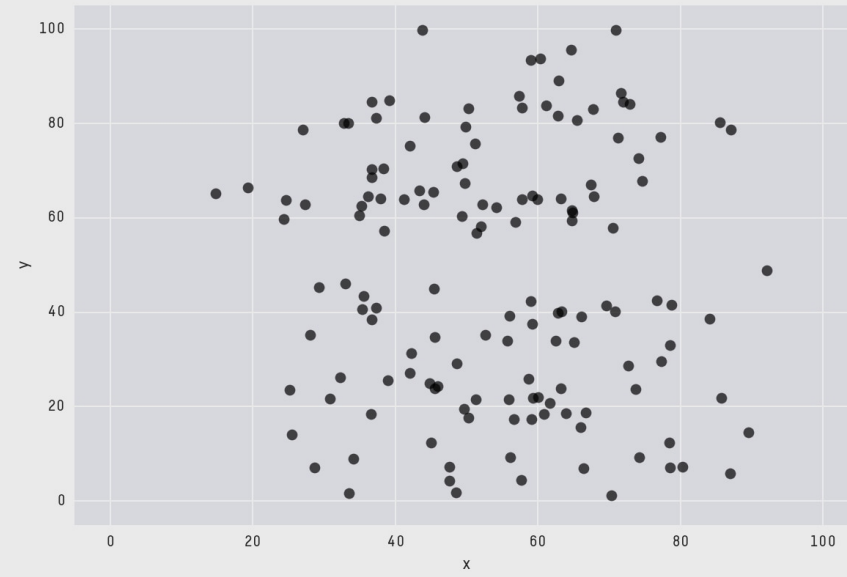
Descriptive Statistics & Summary

“We must suppress some of the truth to communicate the truth... In short, the techniques of descriptive statistics are designed to match the salient features of the data set to human cognitive abilities.”

-I.J. Good (1983)



X Mean: 54.26
 Y Mean: 47.83
 X SD : 16.76
 Y SD : 26.93
 Corr. : -0.06



What do we get from EDA?

Exploratory Data Analysis

- Understand data properties
- Discover Patterns (You get to play detective!)
- Generate & Frame Hypothesis
- Suggest modeling strategies
- Check assumptions (sanity checks)
- Communicate results (present the data)



John Tukey

1915 — 2000

Was maybe the first person to be what we now call a data scientist

Cool things he did:

FFT, box plot, HSD, coined the terms bit and software

From 1977

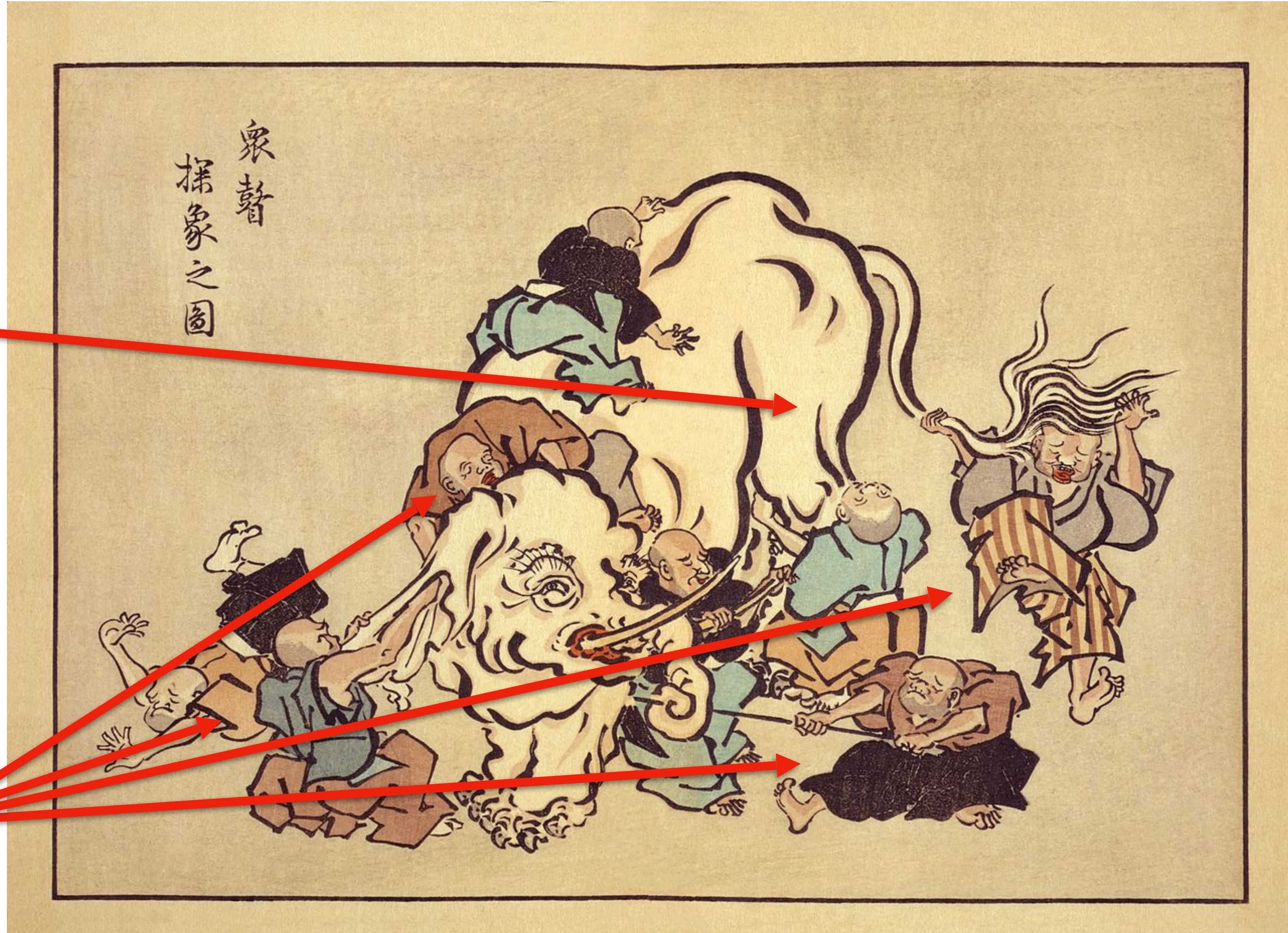
John W. Tukey

EXPLORATORY DATA ANALYSIS



The
dataset

You

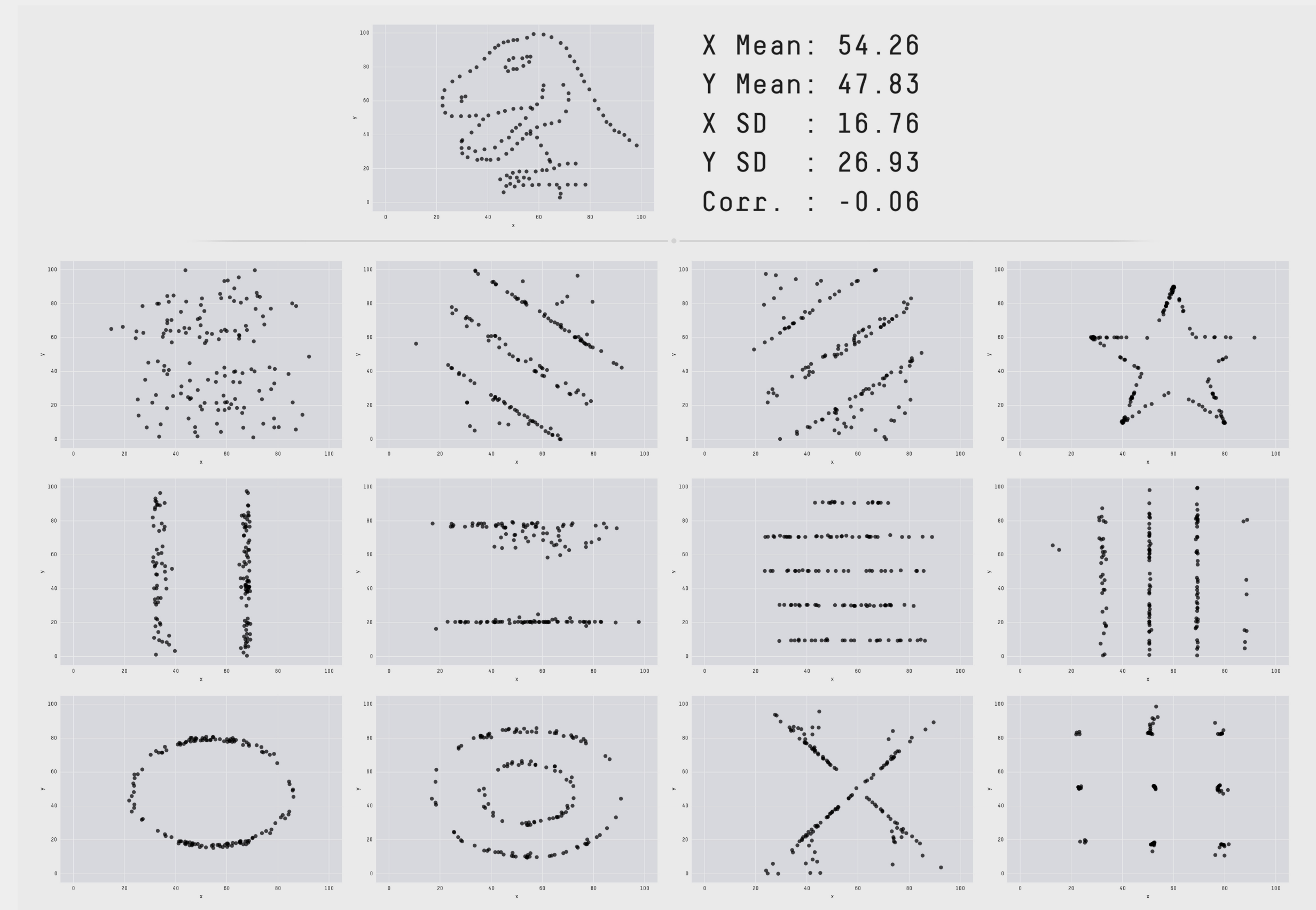


The general principles of EDA

- Generate plots and tables to explore relationships
- Prefer methods that do not assume a distribution (unless you've checked first that the data fit the assumption)
- If necessary, transform variables and generate more plots and tables

Start raw

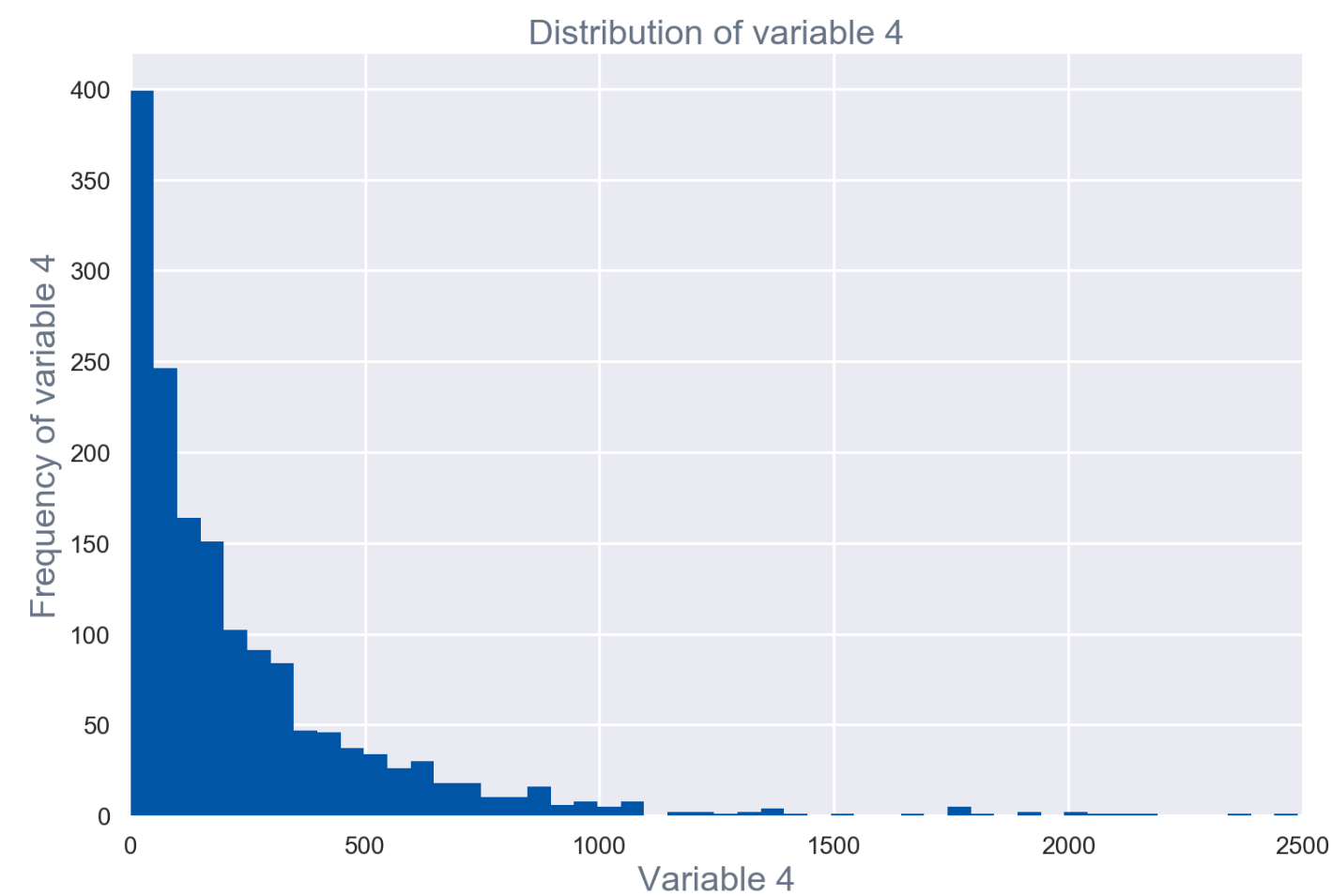
- Examine raw data in the most direct way you can reasonably do so
- View a random sample of the data
- Plots, especially subsets of variables and dimensionality reduction
- Helpful for seeing weirdness, missingness, outliers, min/max/typical values



Exploratory

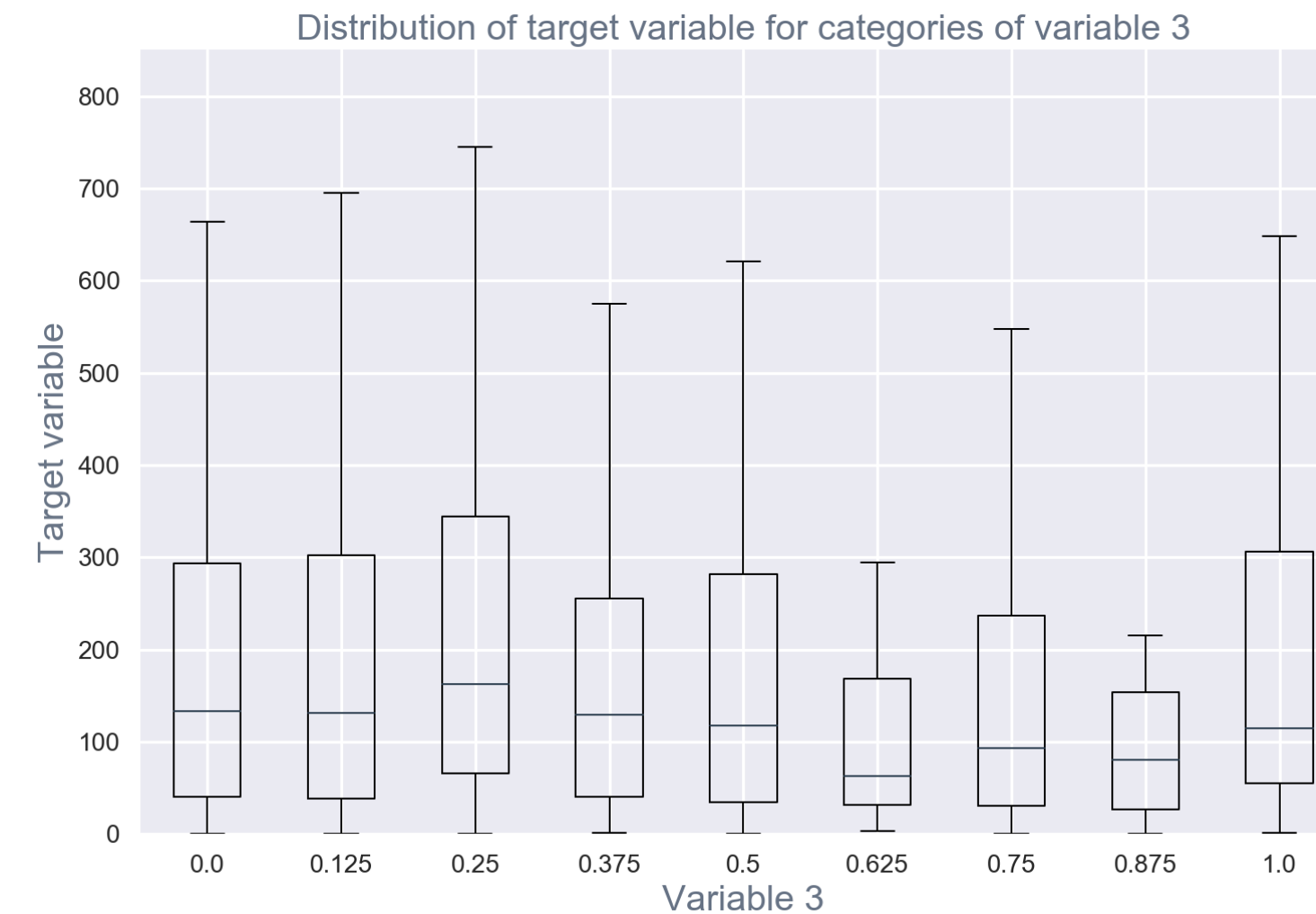
EDA Approaches to “Get a Feel for the Data”

Understanding the relationship between variables in your dataset



Univariate

understanding a single variable
i.e.: histogram, densityplot, barplot



Bivariate

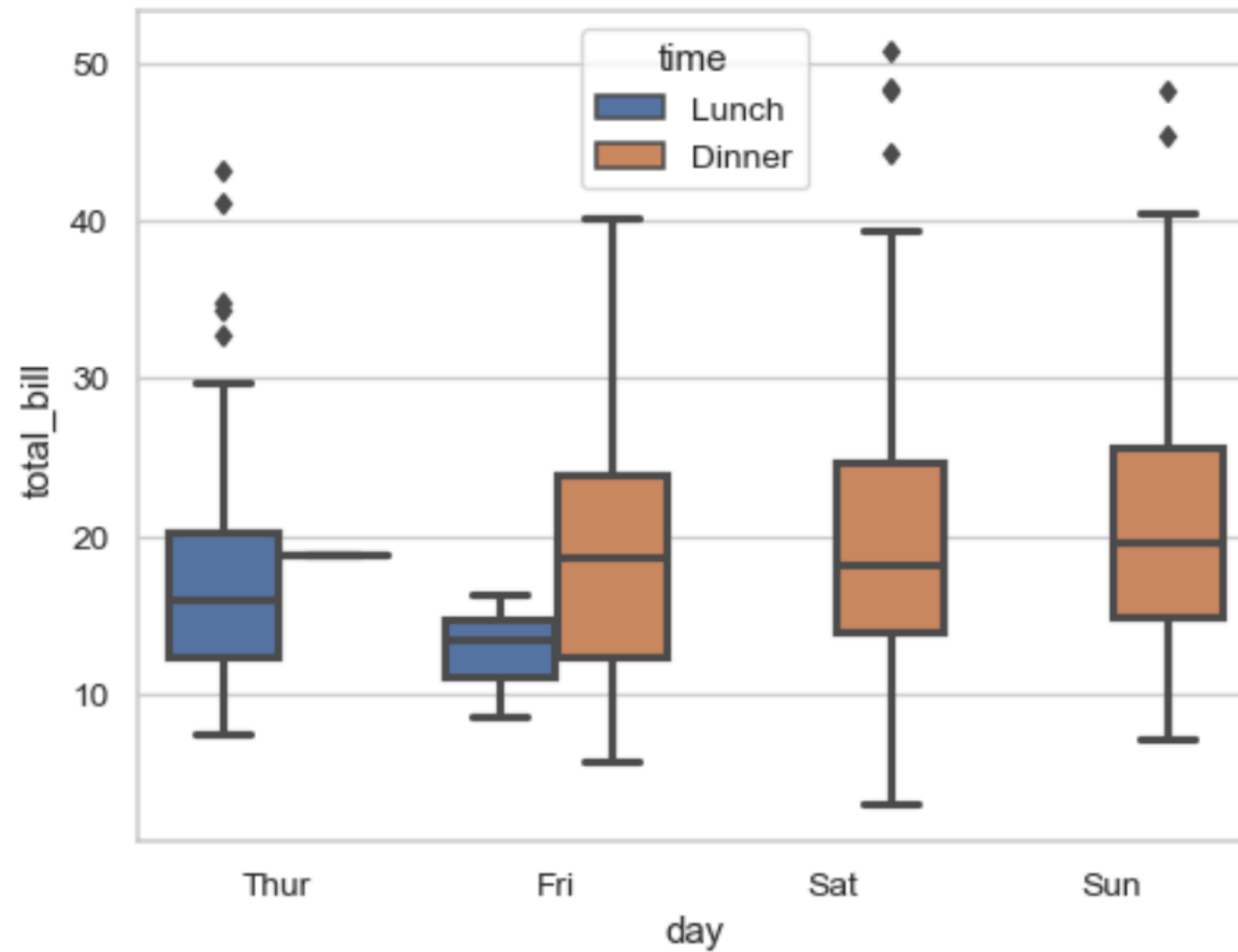
understanding relationship between 2
variables
i.e.: boxplot, scatterplot, grouped
barplot, boxplot



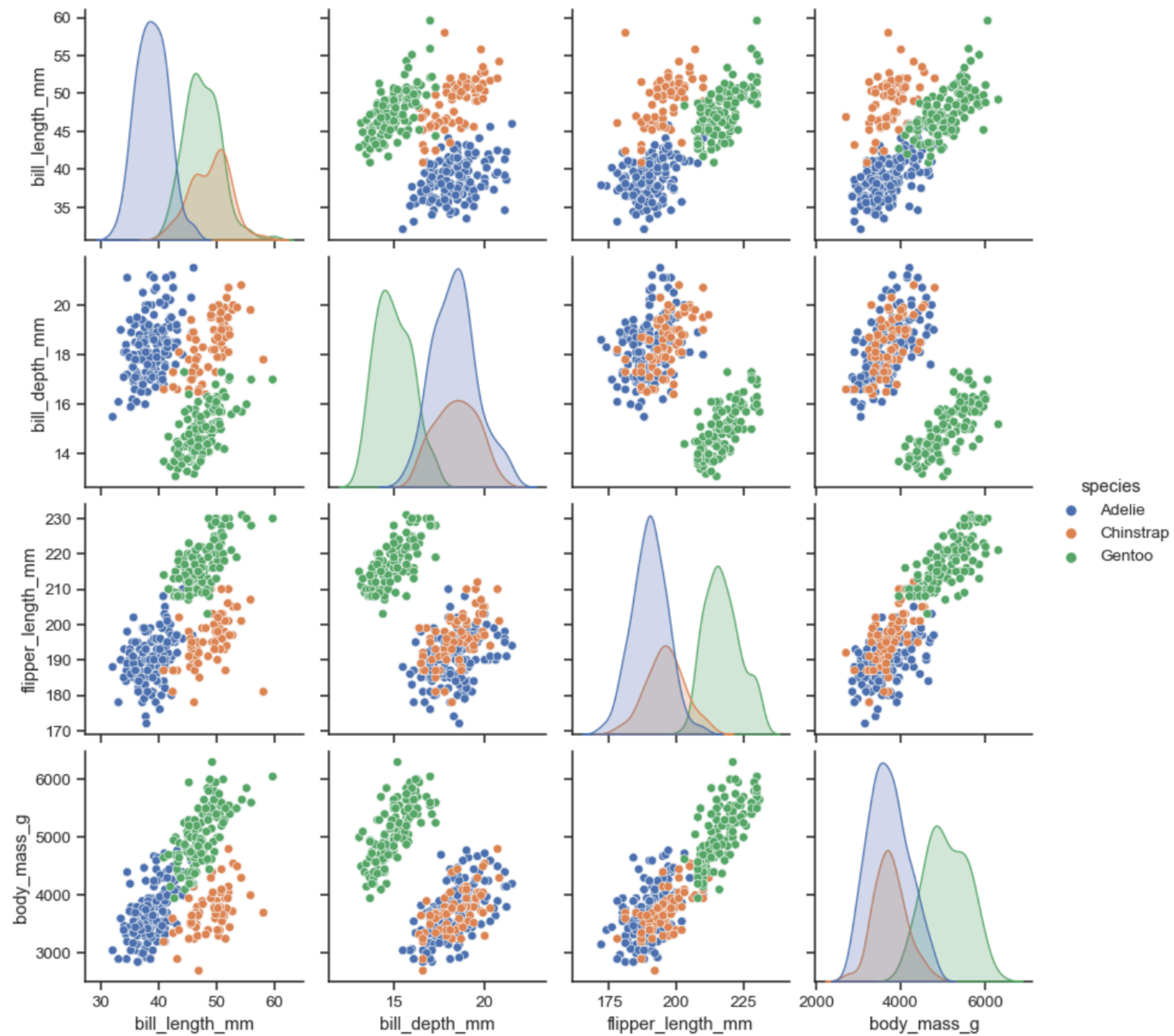
Dimensionality Reduction

projecting high-D data into a lower-
D space
i.e.: PCA, ICA, Clustering

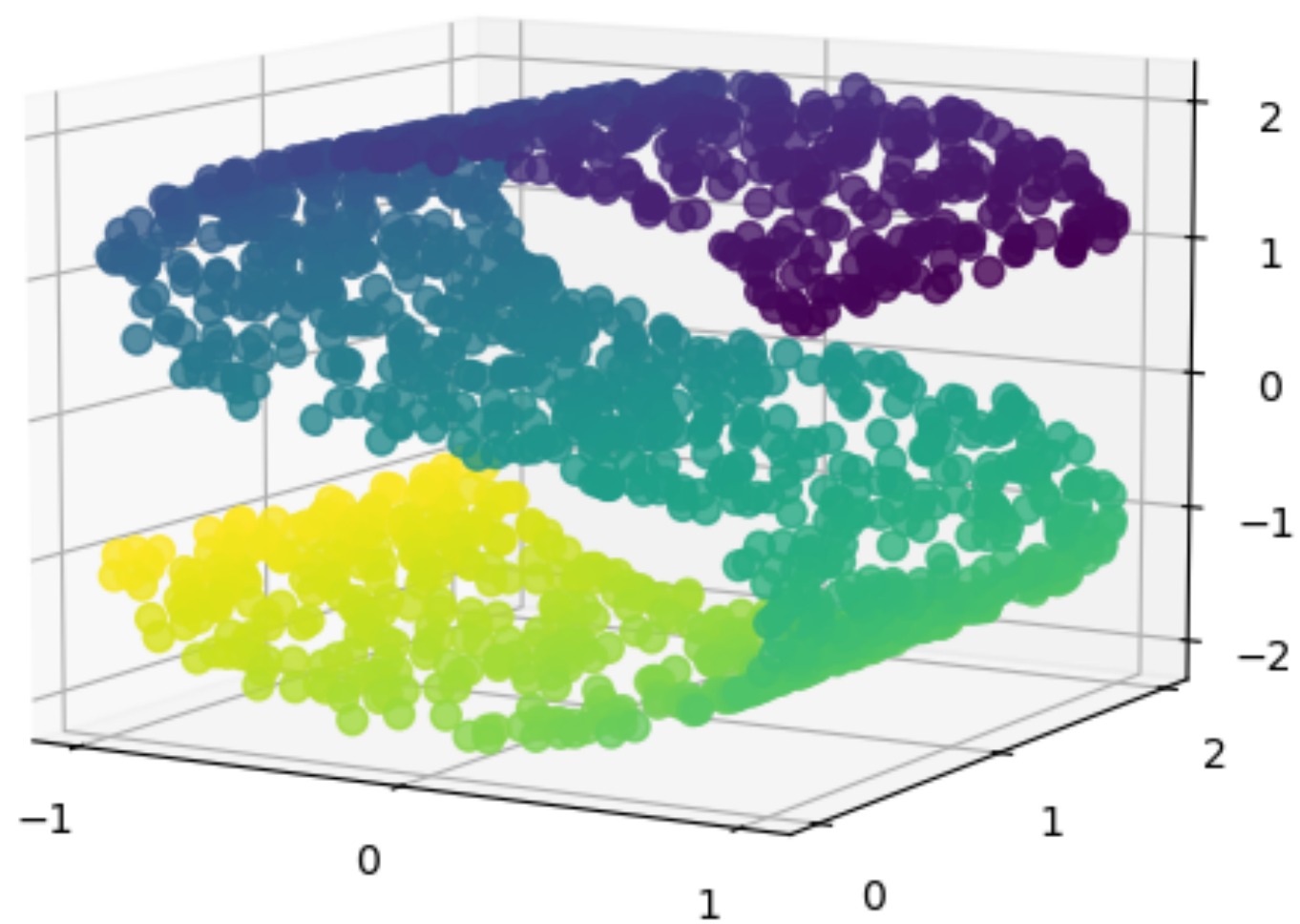

```
>>> ax = sns.boxplot(x="day", y="total_bill", hue="time",  
...                   data=tips, linewidth=2.5)
```



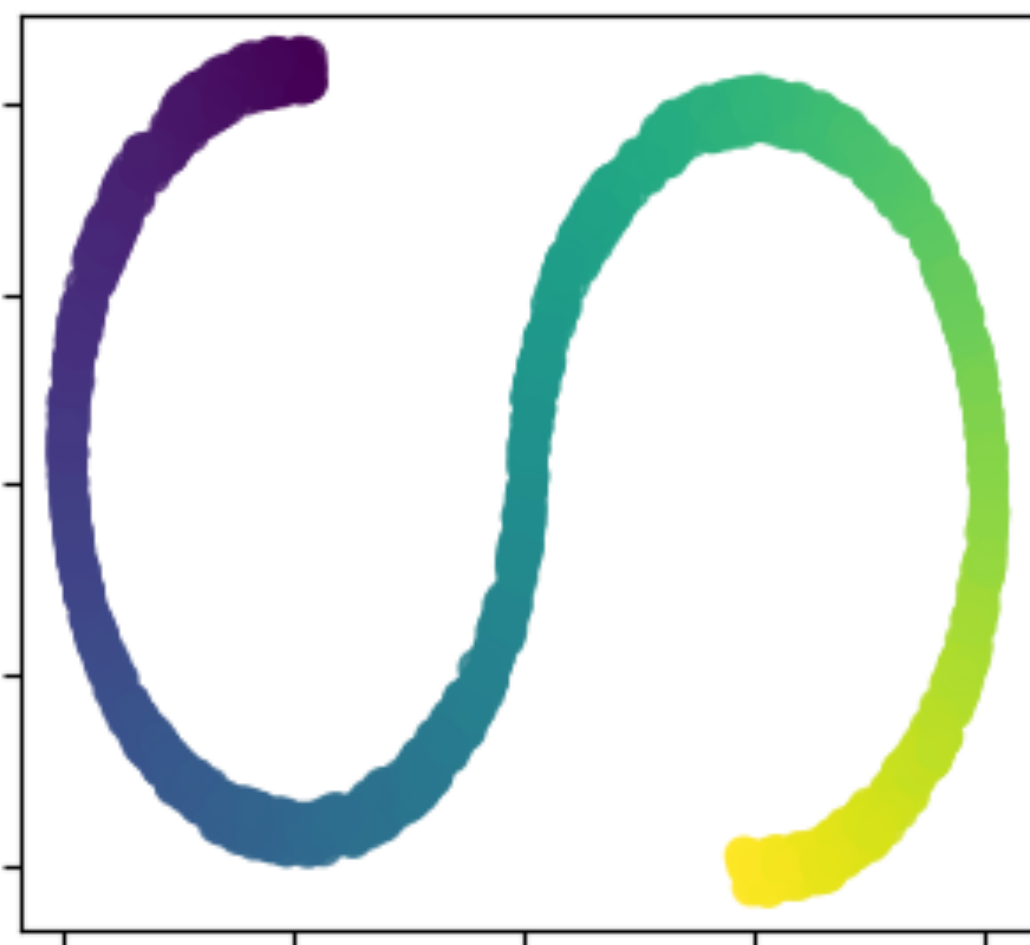

```
sns.pairplot(penguins, hue="species")
```



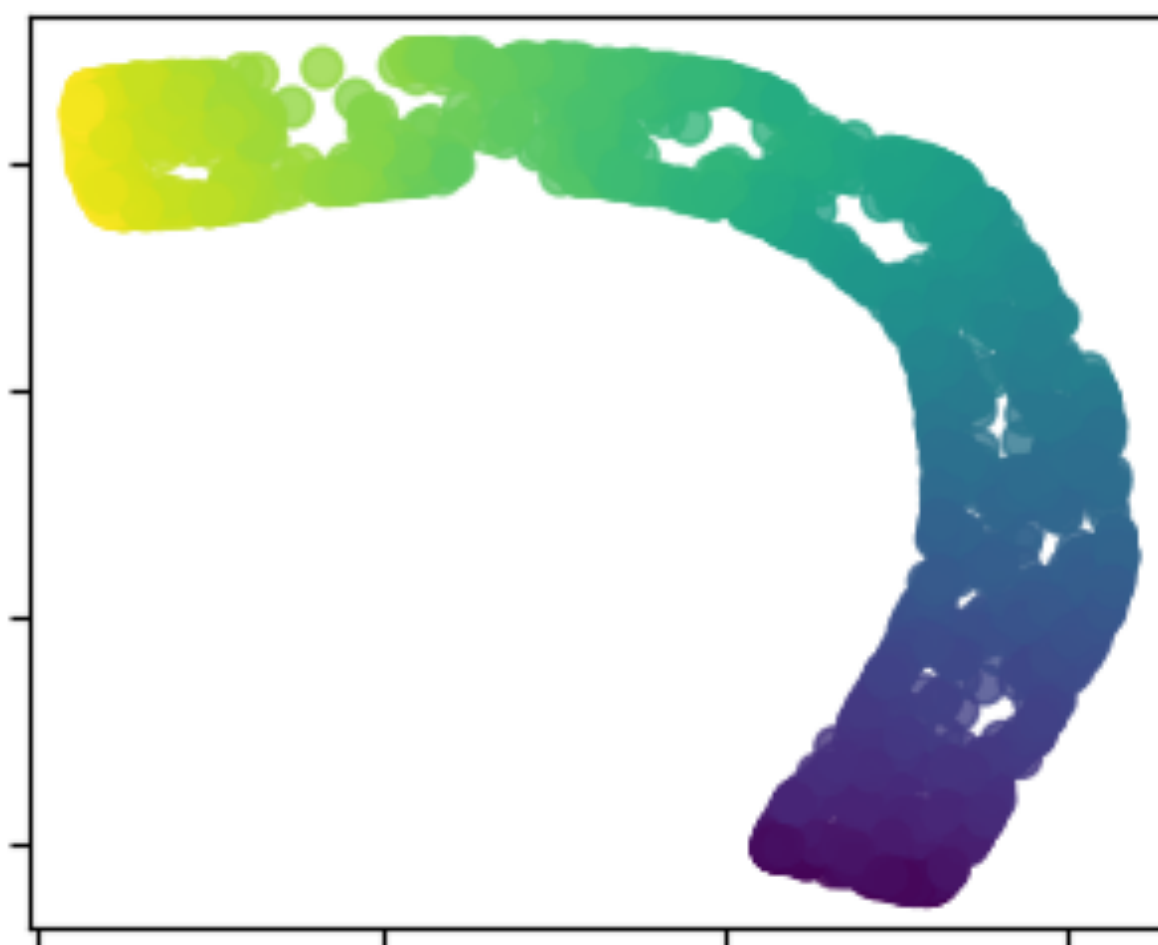
3D S-curve dataset



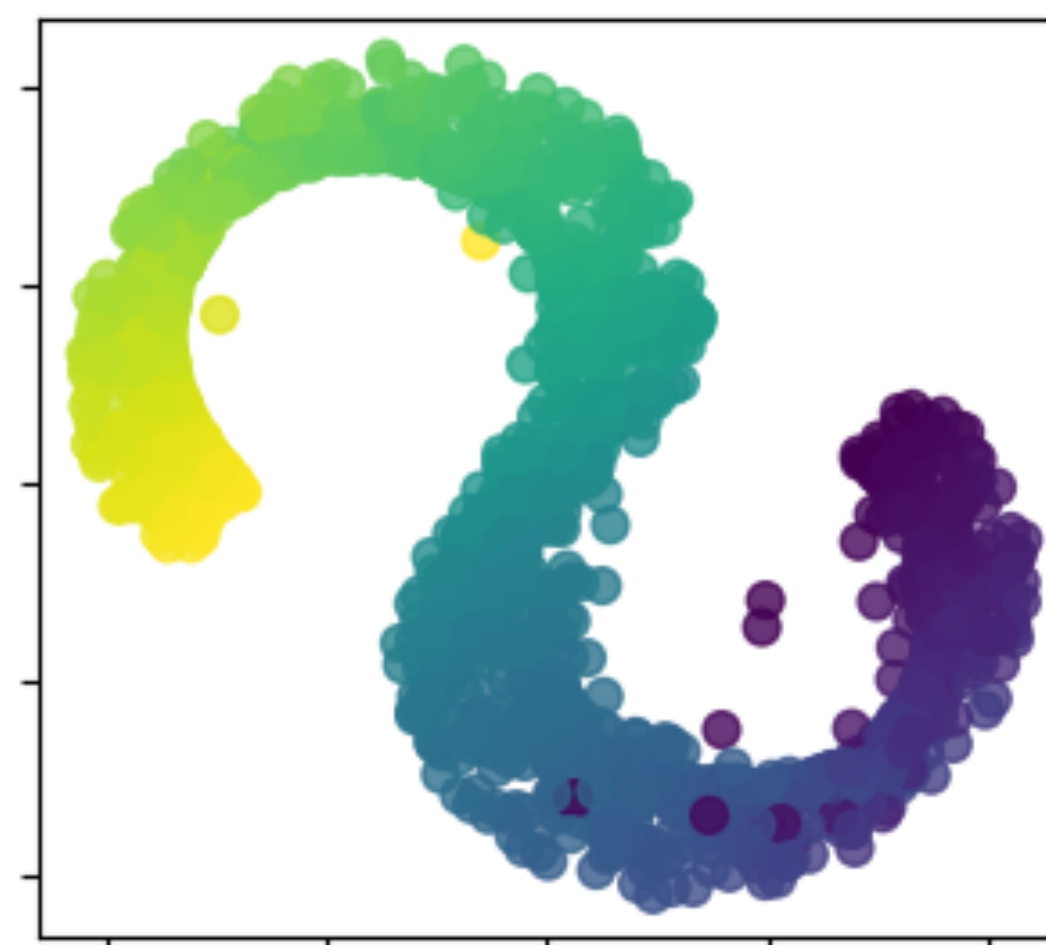
PCA

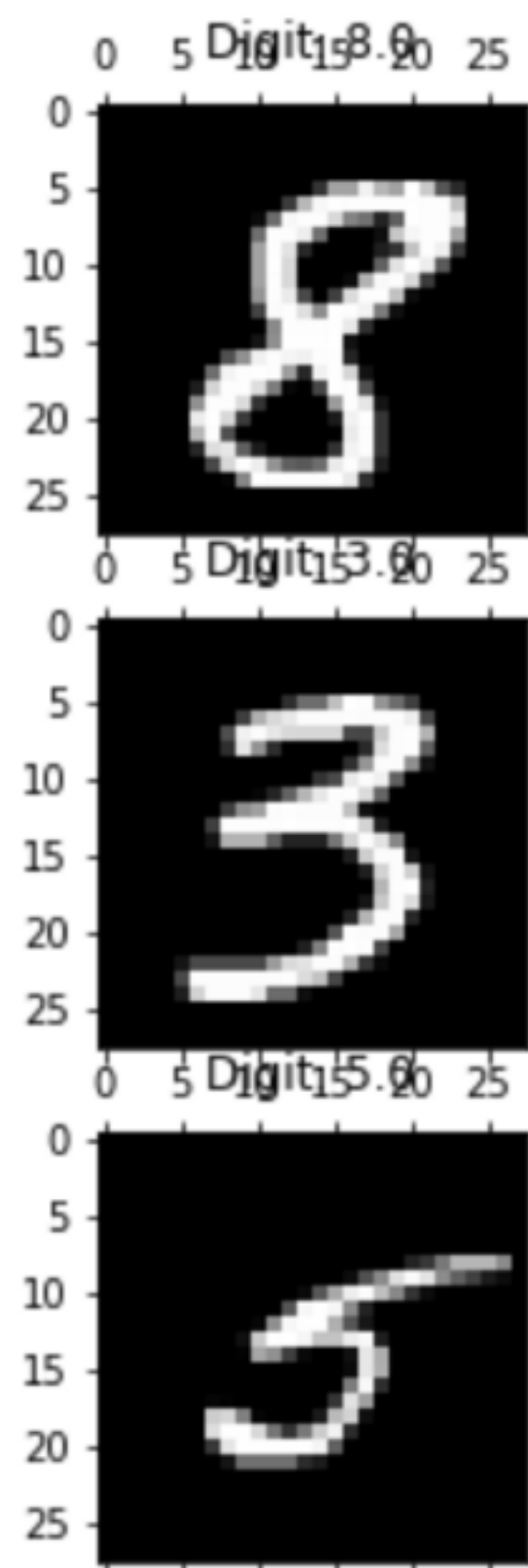
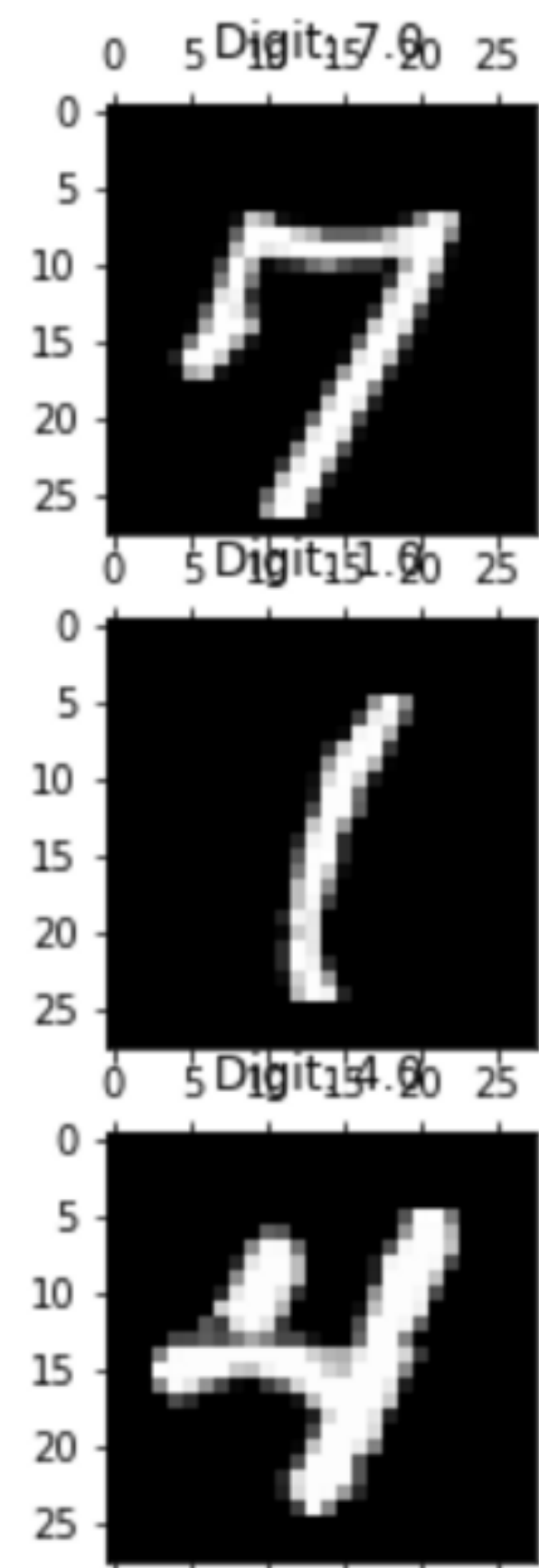


T-distributed Stochastic Neighbor Embedding

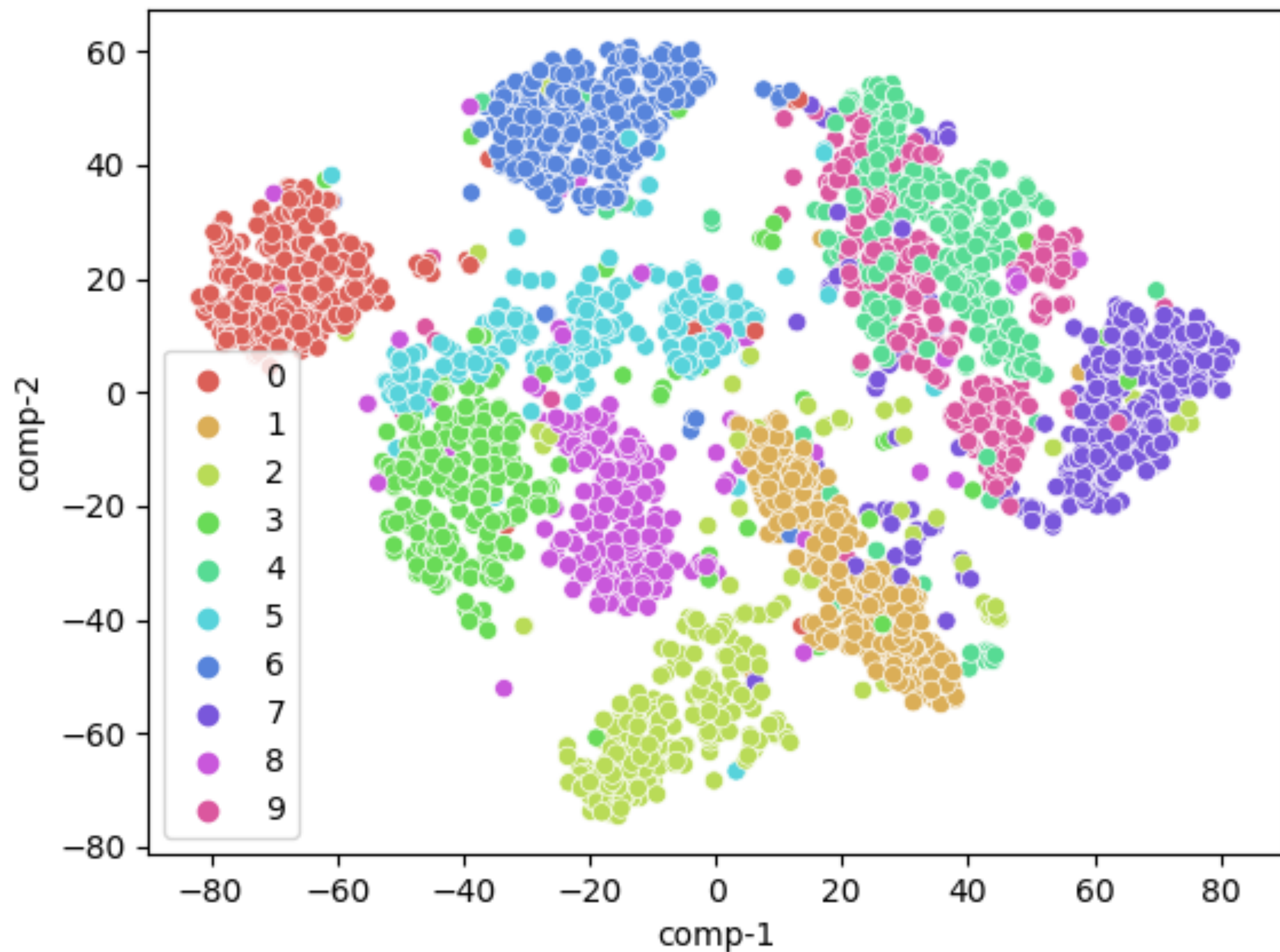


Multidimensional scaling





MNIST data T-SNE projection



Geospatial viz as EDA

How a coastline 100 million years ago influences modern election results in Alabama

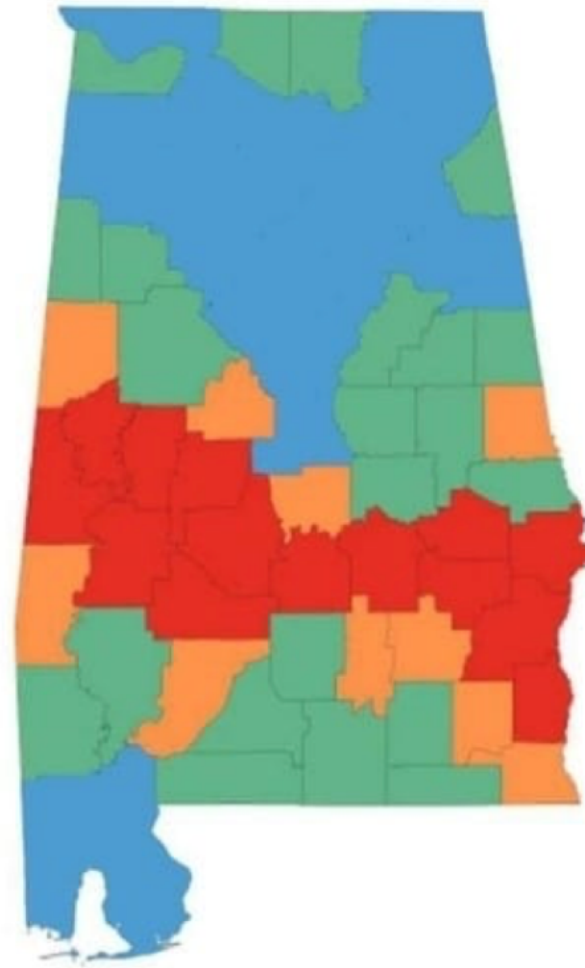
Cretaceous Sediments



Fertile Blackland
Prairie Soil



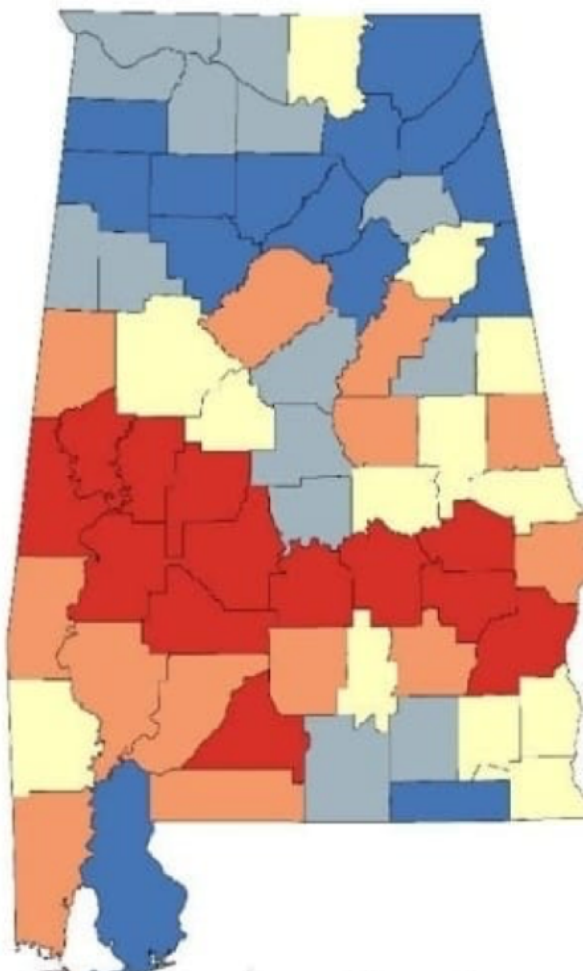
Average Farm Size, 1997



Slave Population, 1860

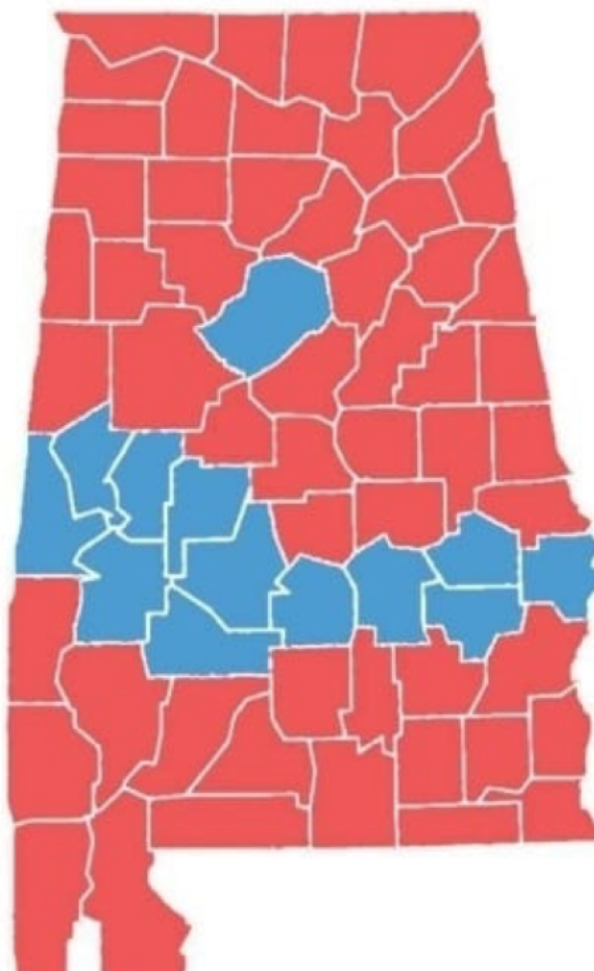


Black population, 2010



-Starkey Comics

Election Results, 2020



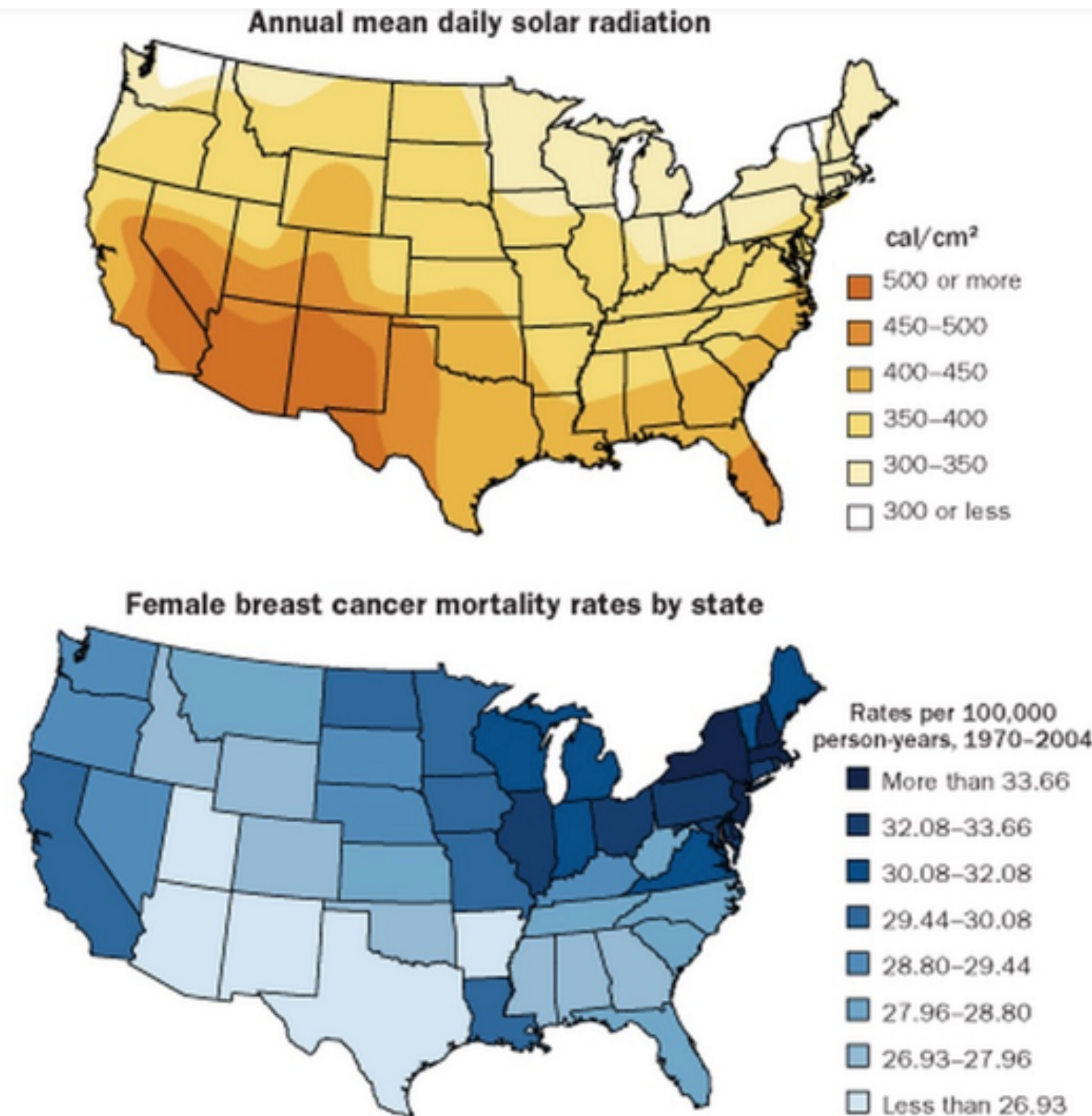
Why Geospatial Analysis?

‘Everything is related to everything else, but near things are more related than distant things.’
(Space > Time)
— Tobler 1979

Why Geospatial Analysis?

“...the purpose of geographic inquiry is to examine relationships between geographic features collectively and to use the relationships to describe the real-world phenomena that map features represent”

—Clarke 2001

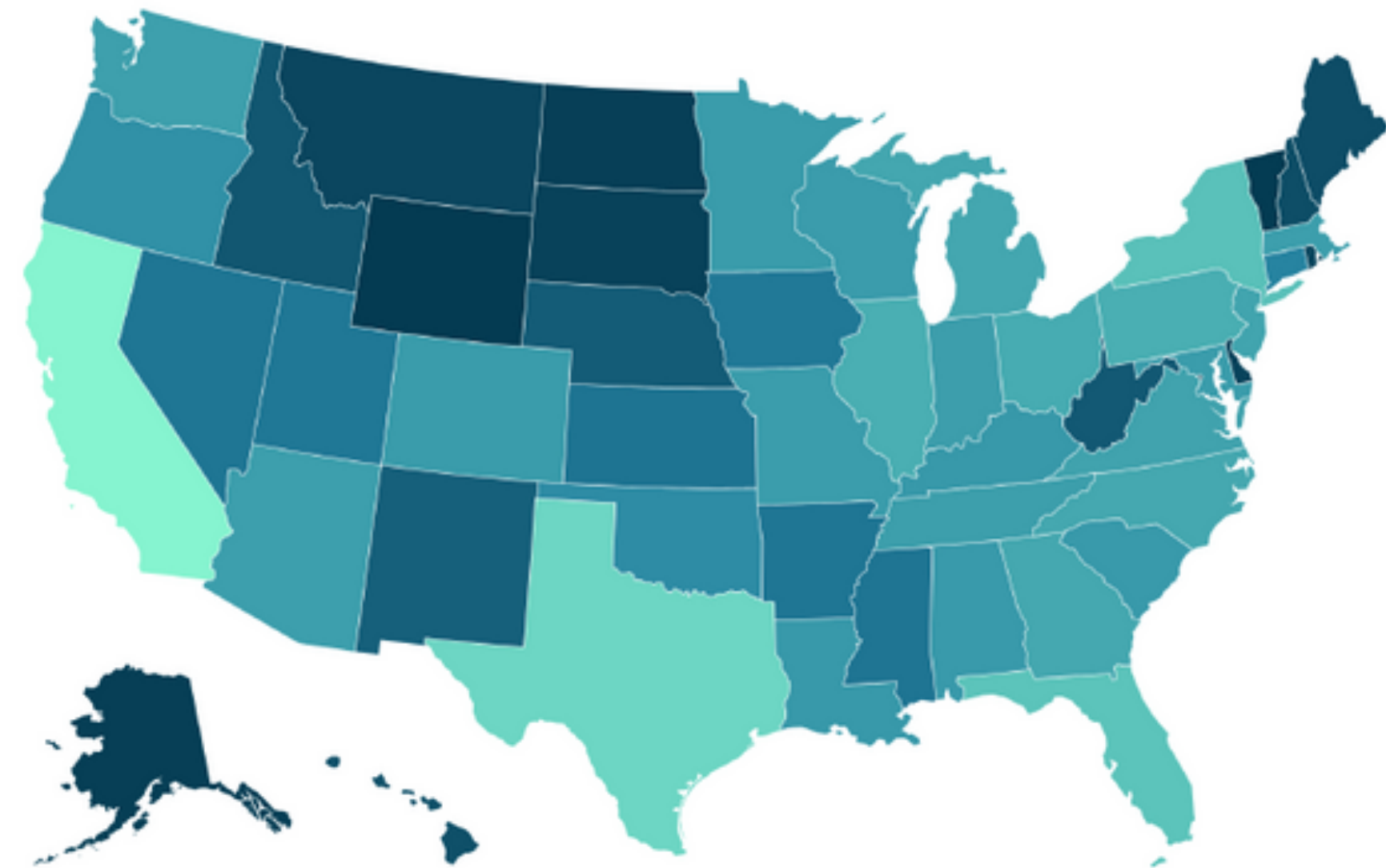


ON THE MAP Scientists who study vitamin D can't help but notice that a host of diseases seem to vary with latitude. Type 1 diabetes, multiple sclerosis and even some cancers appear to be more common in areas that get less sun -- meaning less opportunity for the body to produce vitamin D. The maps above illustrate the apparent link between solar radiation and breast cancer mortality rates.

SOURCE, FROM TOP: D. M. HARRIS AND V.L.W. GO // J. OF NUTRITION 2004; NATIONAL CANCER INSTITUTE

Choropleth maps are useful for visualizing *clear regional patterns* in the data

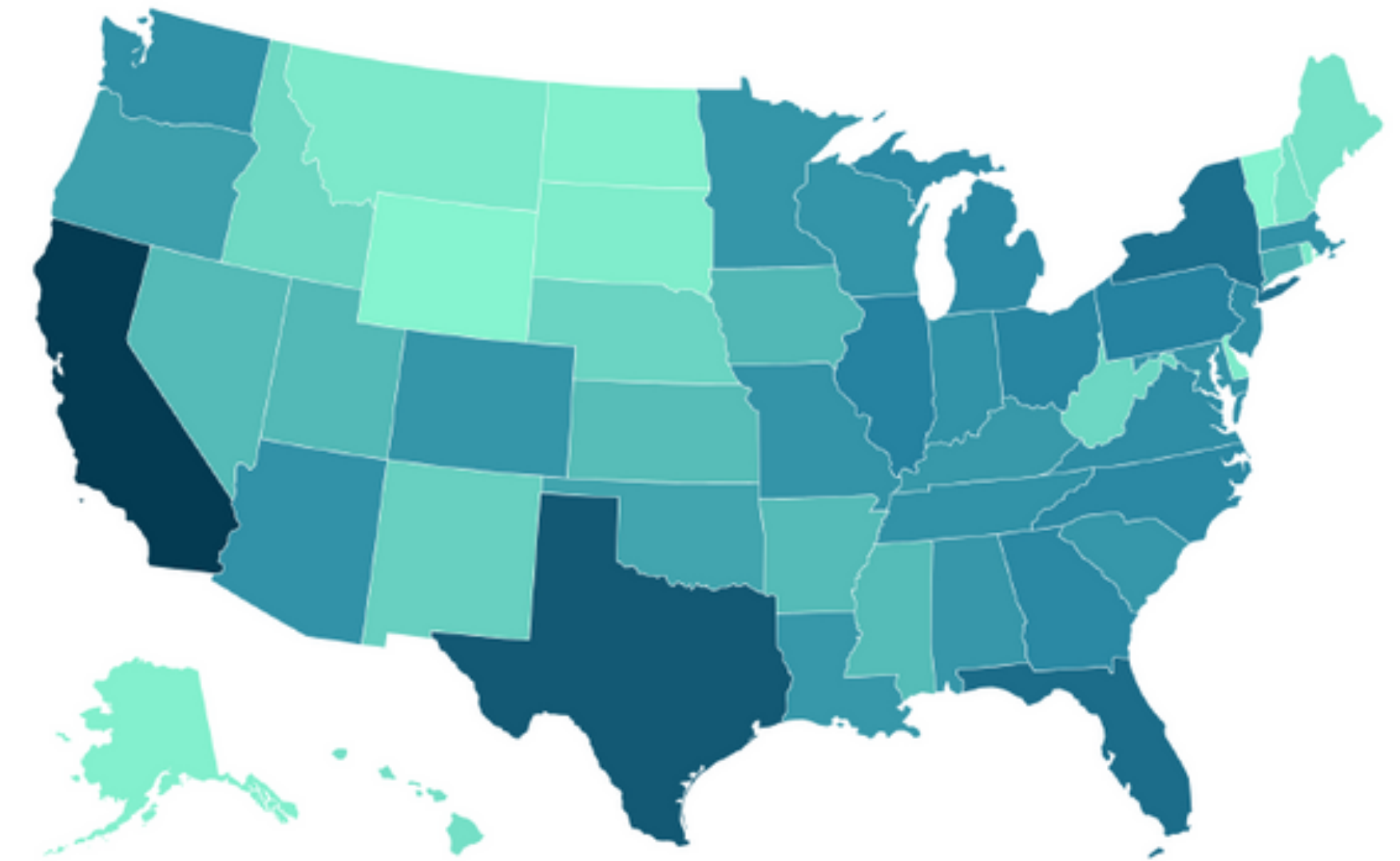
NOT IDEAL



LOW POPULATION HIGH



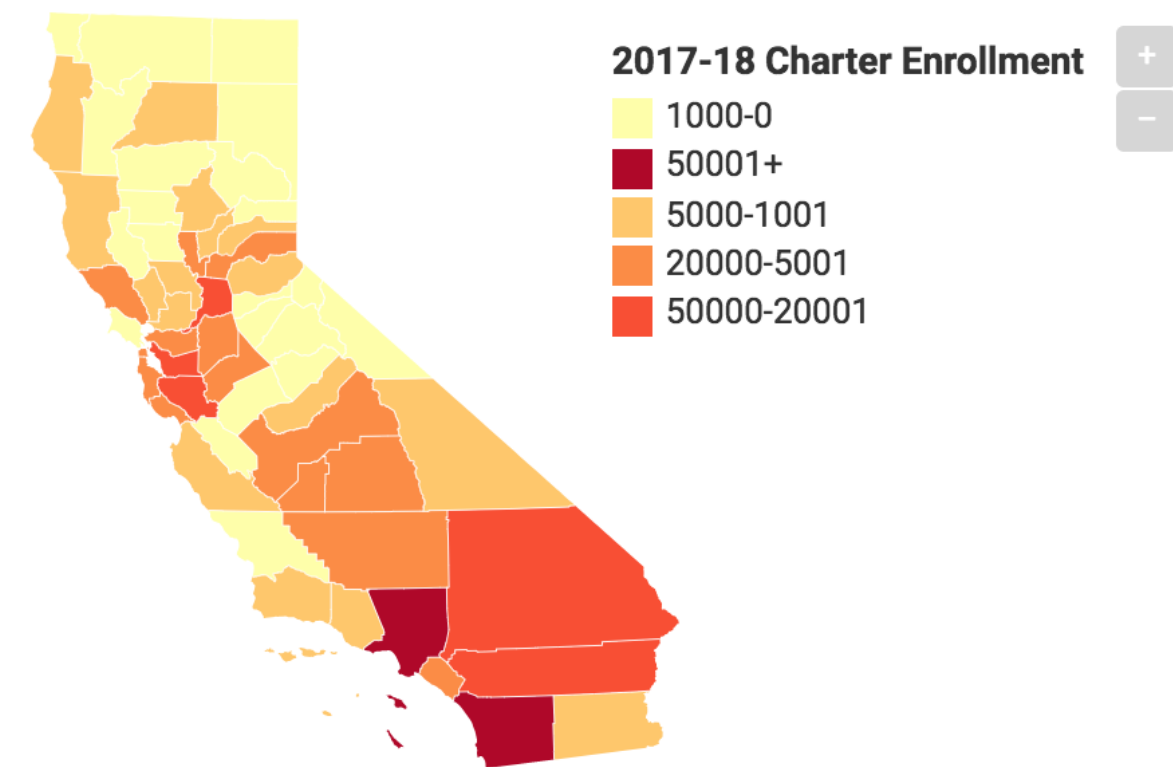
BETTER



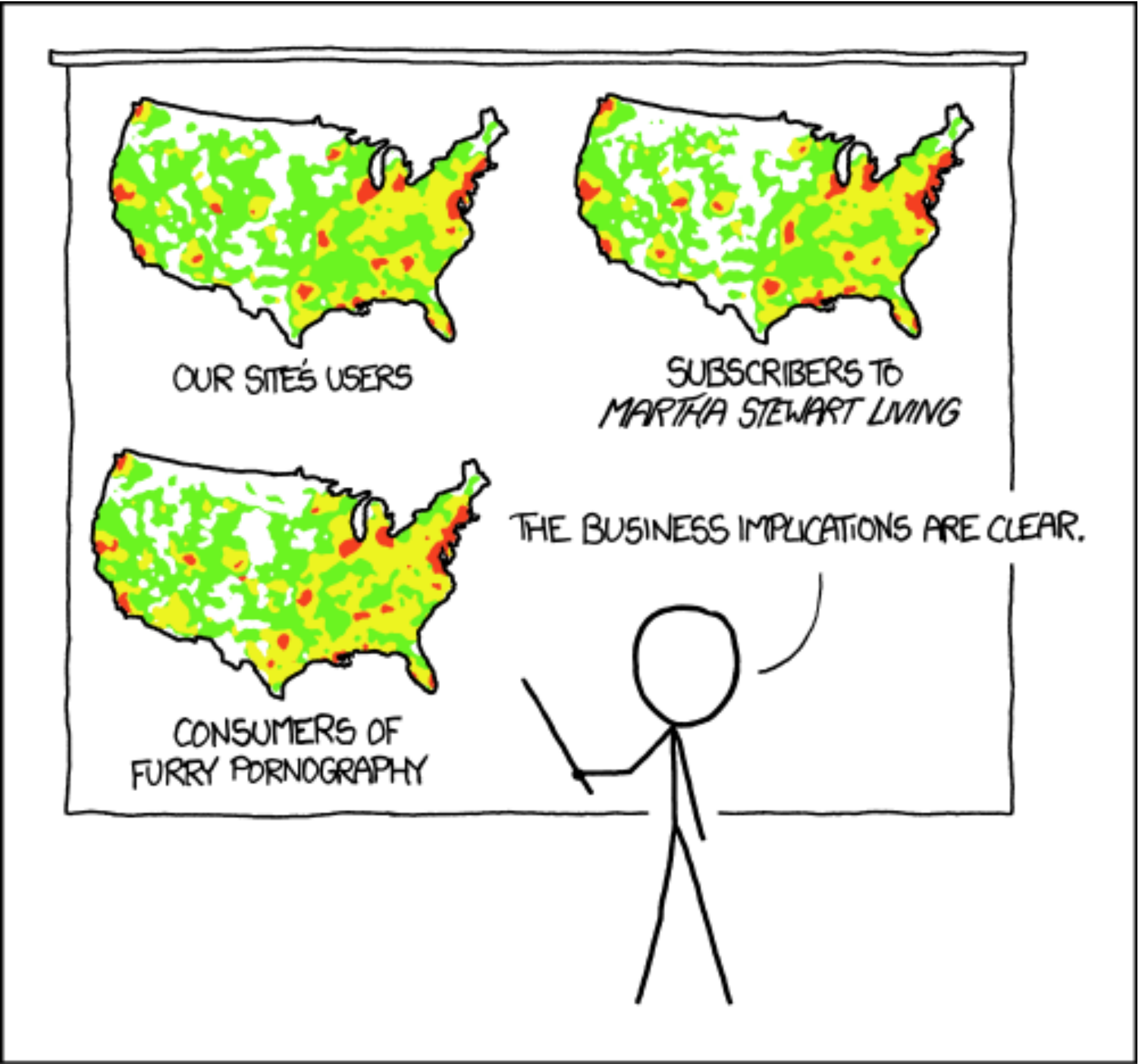
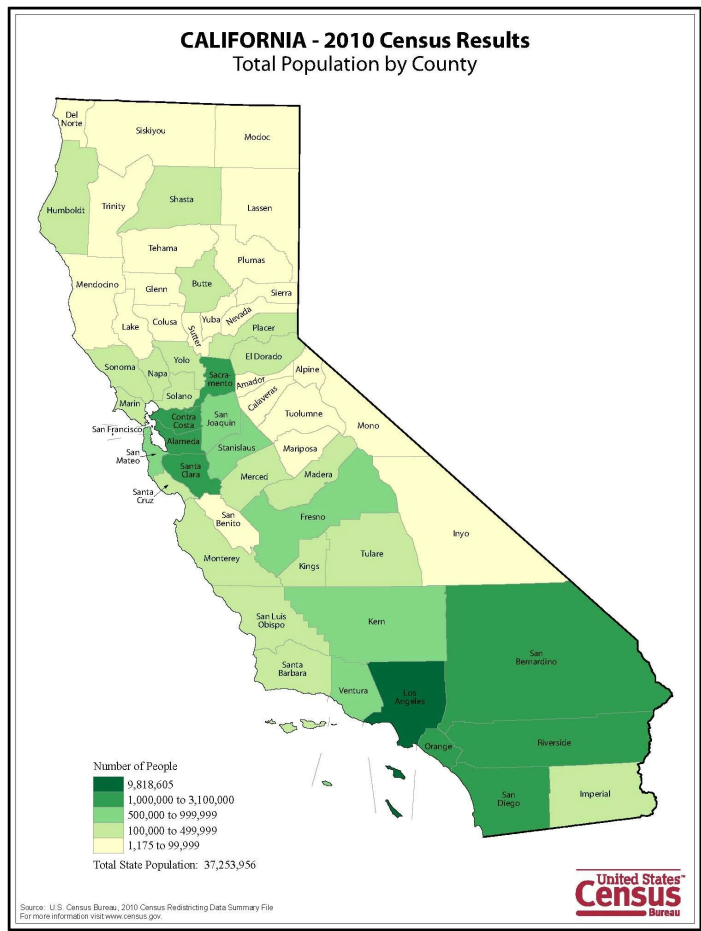
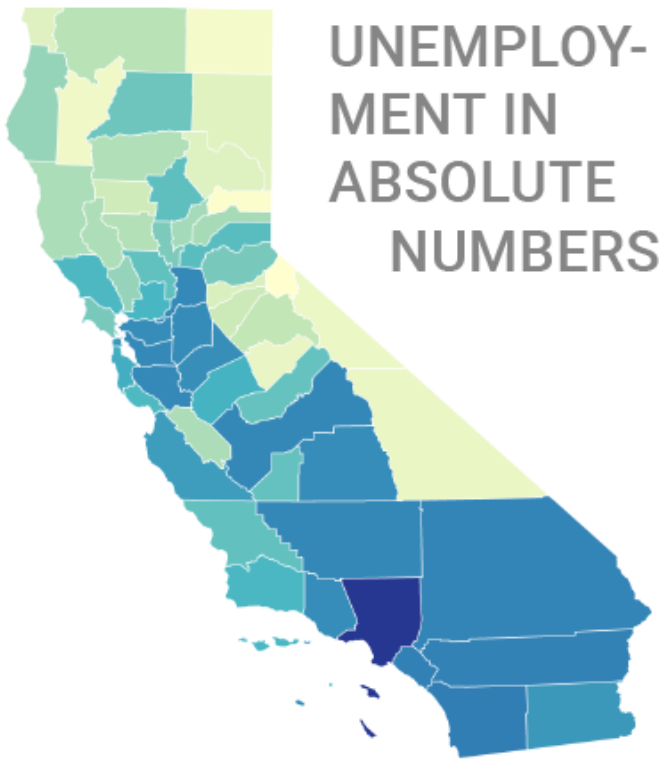
LOW POPULATION HIGH



Use light colors for low values. Dark colors for high values.



Map: Ricardo Cano • Source: California Department of Education • [Get the data](#) • Created with Datawrapper

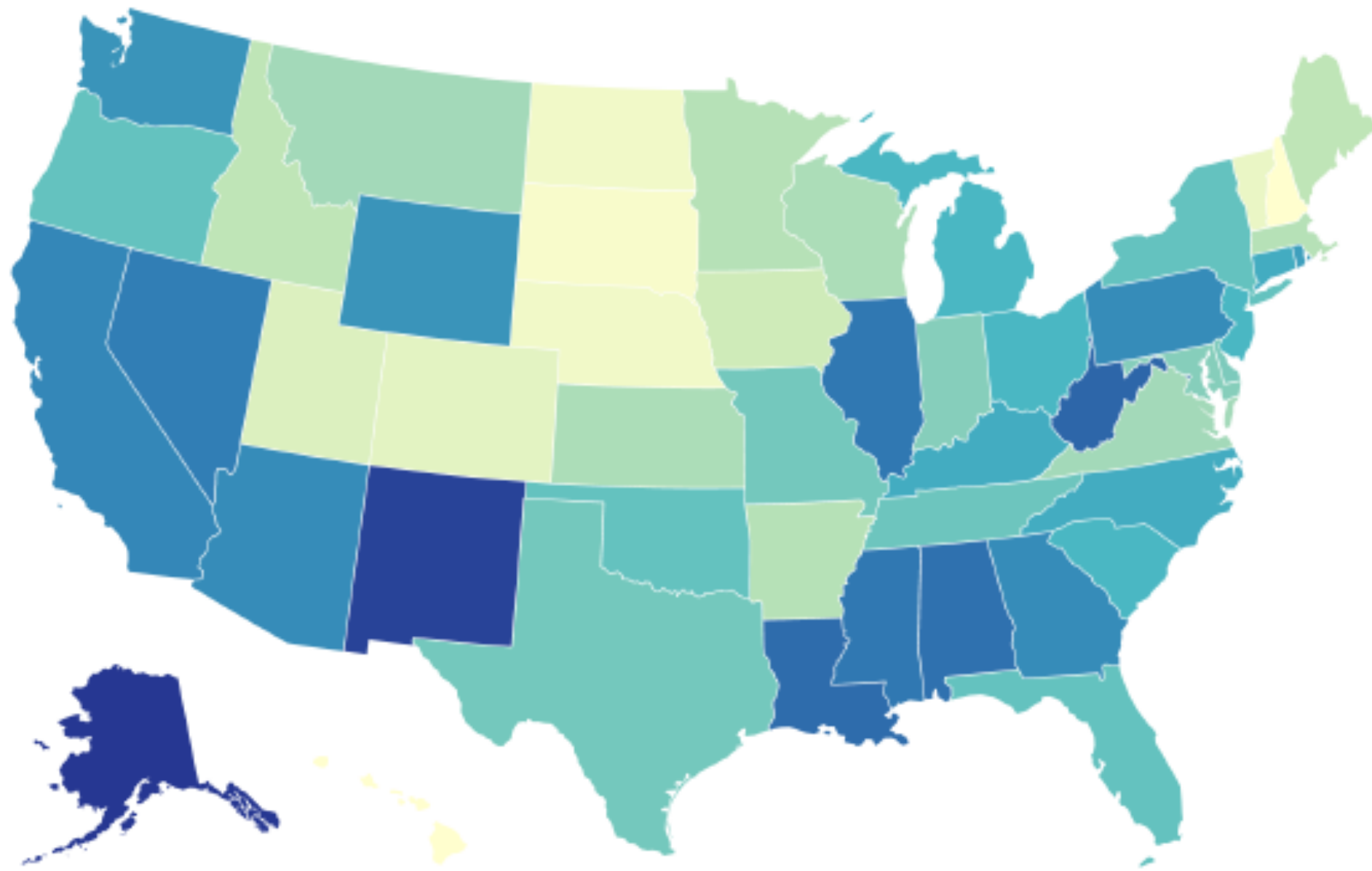


PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS

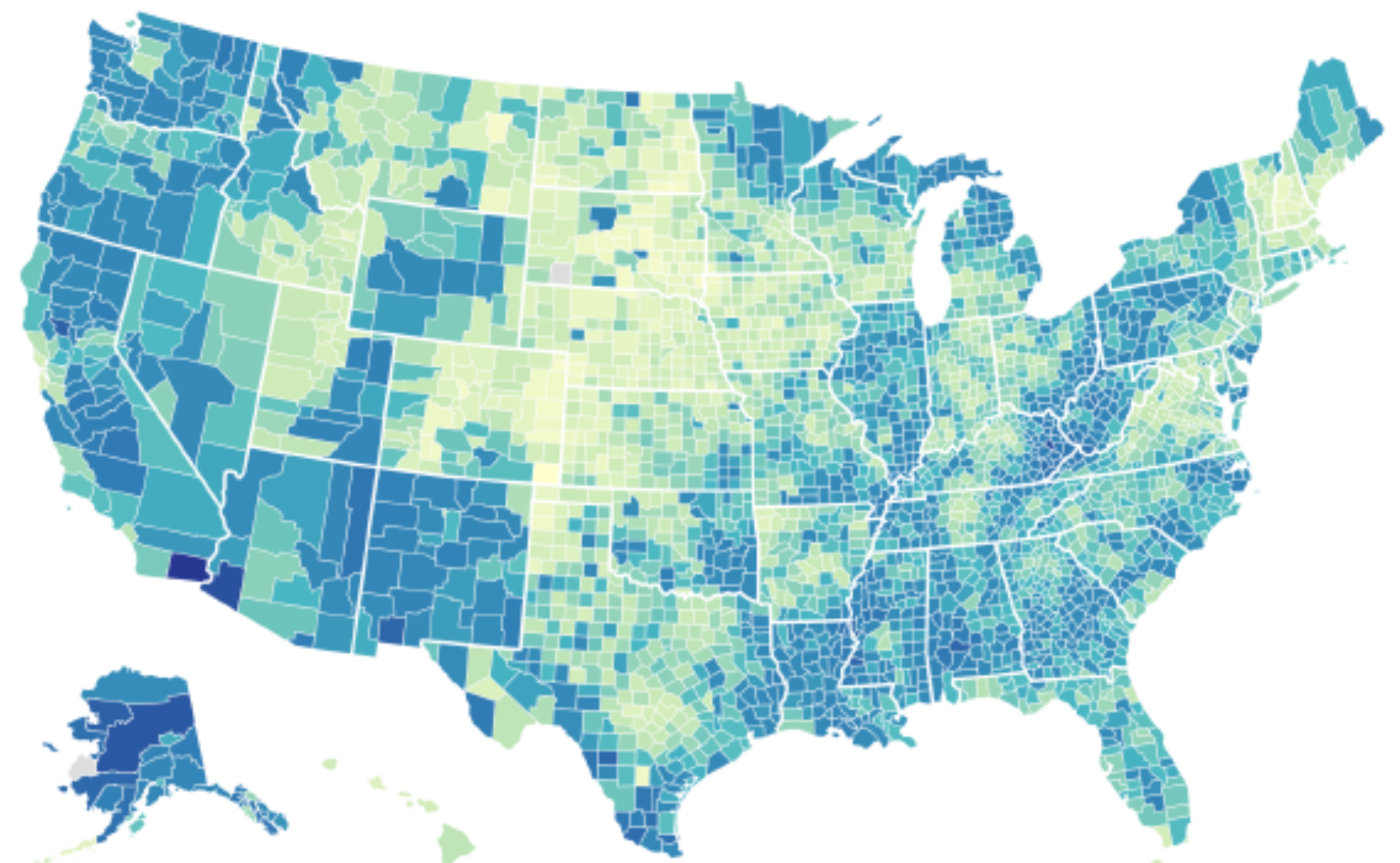
If you don't normalize the statistic
(divide by population or by area)
choropleth maps can be misleading

Consider using the smallest unit possible
(but there are exceptions!)

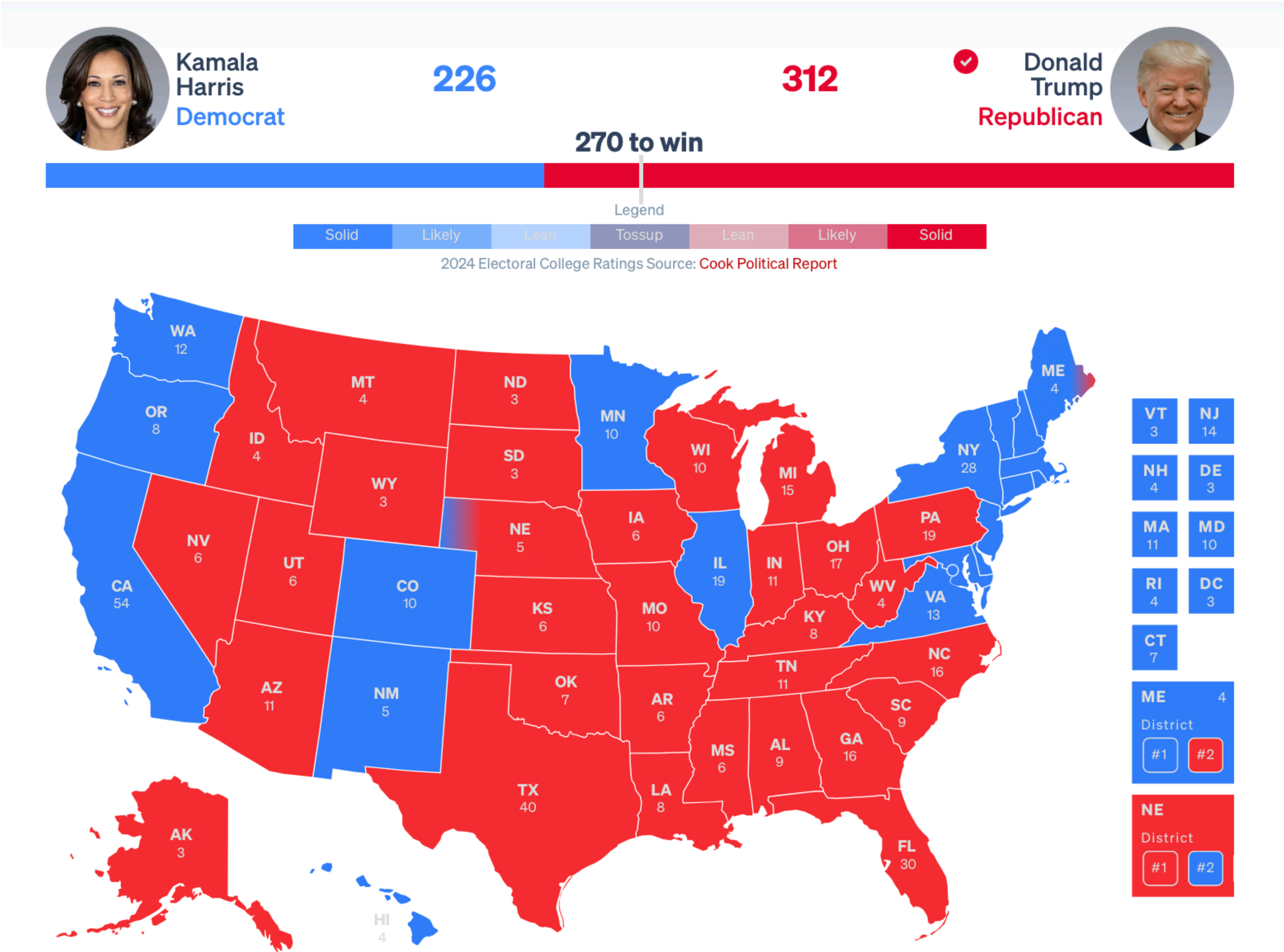
NOT IDEAL



BETTER

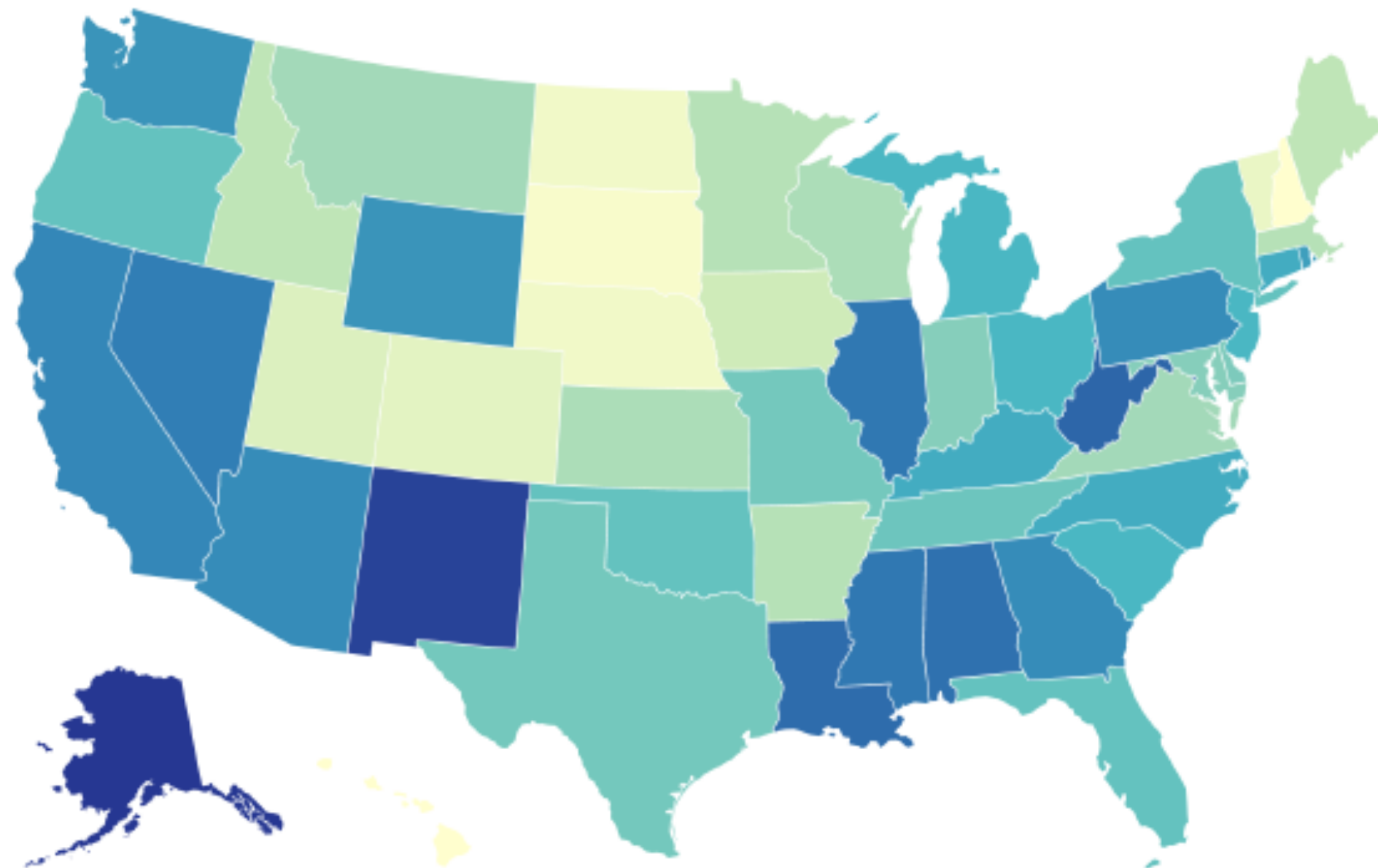


Sometimes summarizing at the state level is OK



Cartograms should be considered when displaying how many people were affected

NOT IDEAL



Choropleths answer “How much area was affected?”

BETTER



Cartograms answer “How many people were affected?”



Kamala Harris
226

✓ Donald Trump
312



270

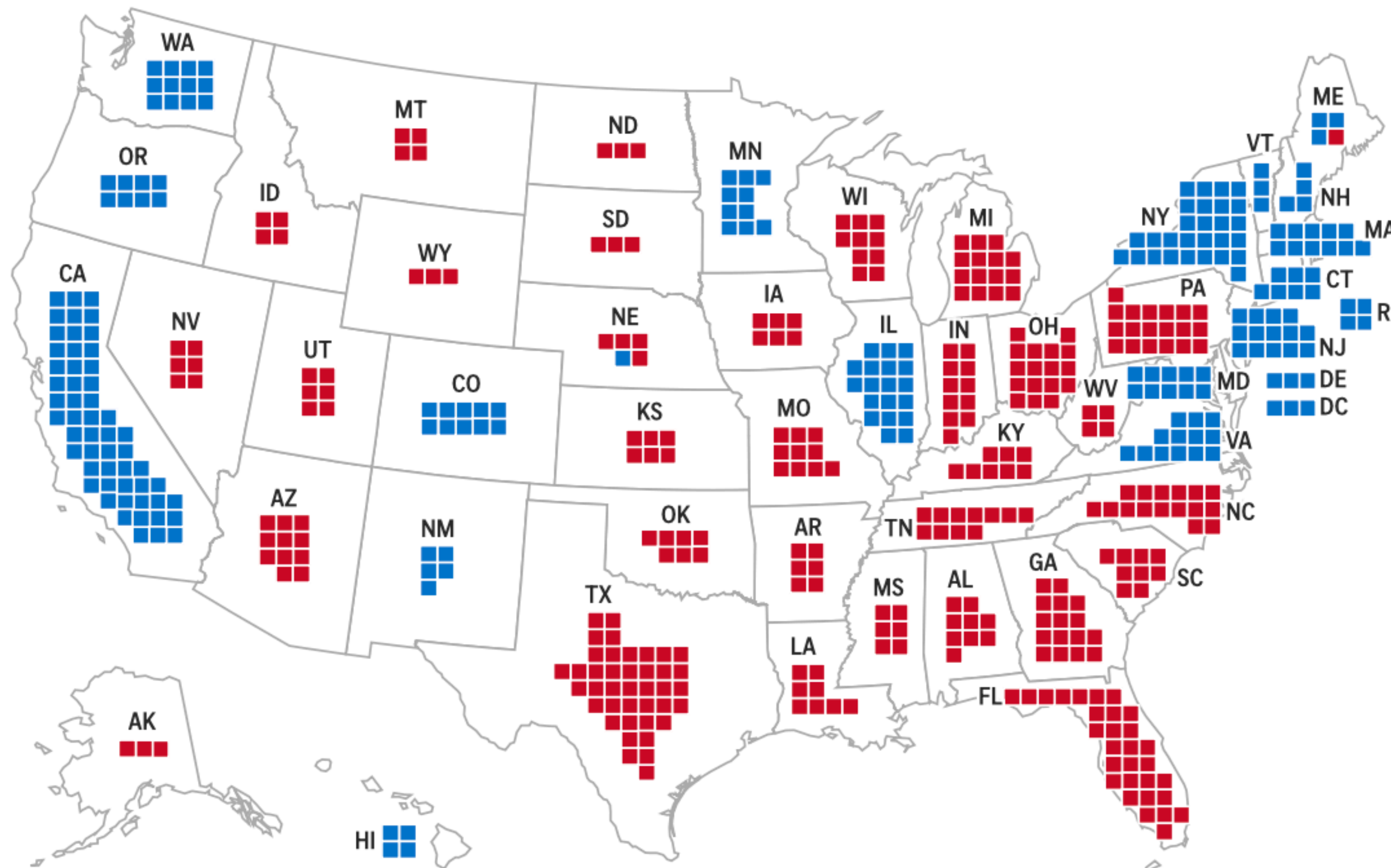


75,019,617 votes (48.4%)

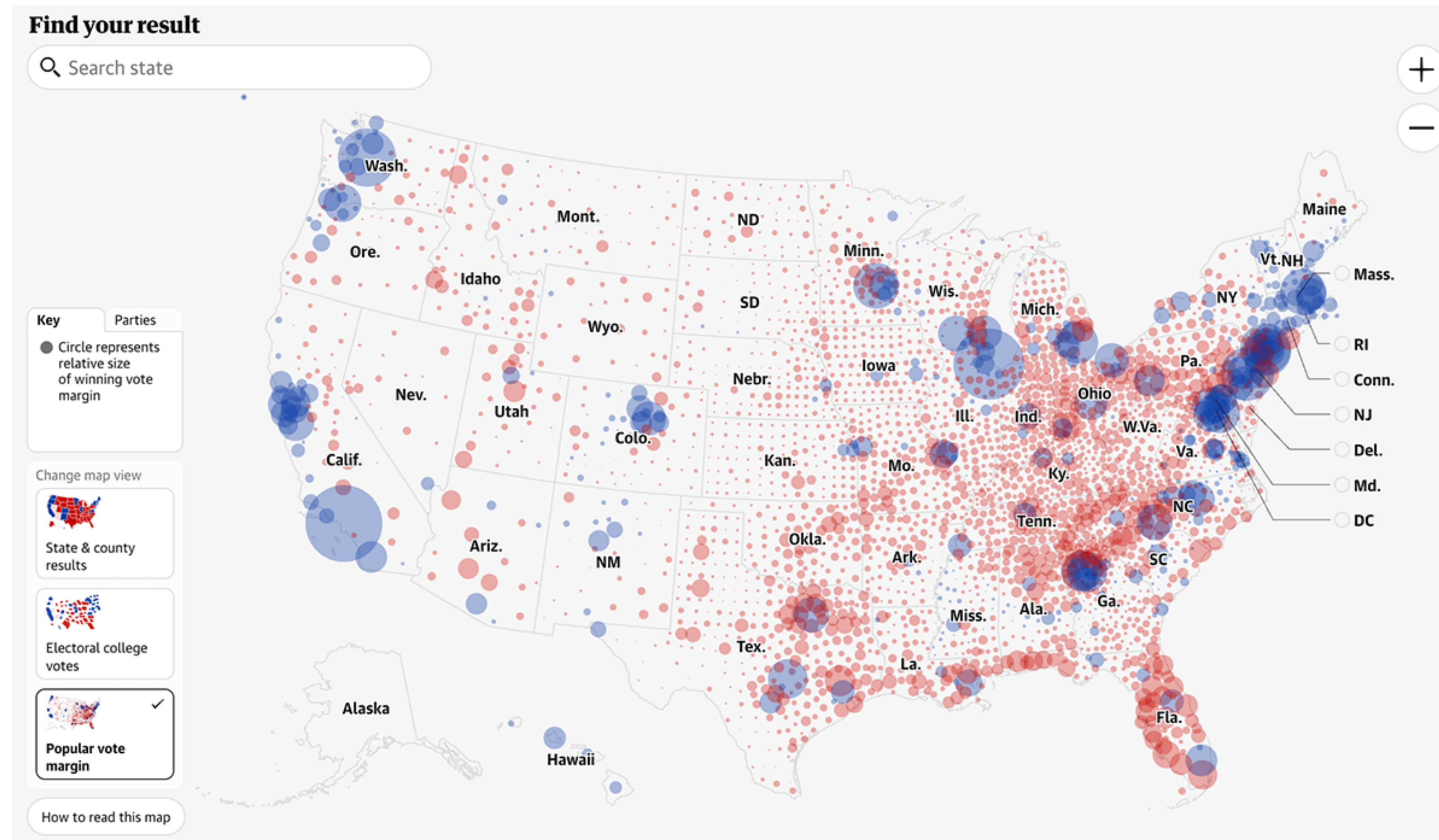
77,304,184 votes (49.9%)

2024 Presidential Election

National Map ▼



This **cartogram** more accurately shows how the populace is split since the number of tiles reflects state populations

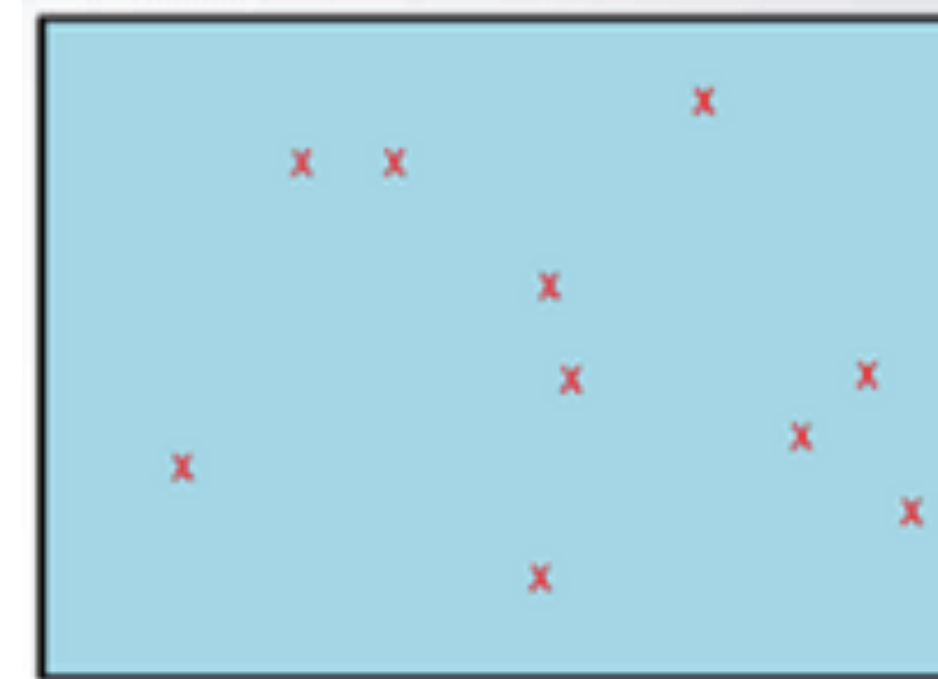


This county level **bubble graph** more accurately tells the full story, since the size of the bubbles reflects population while color shows the political split in that county

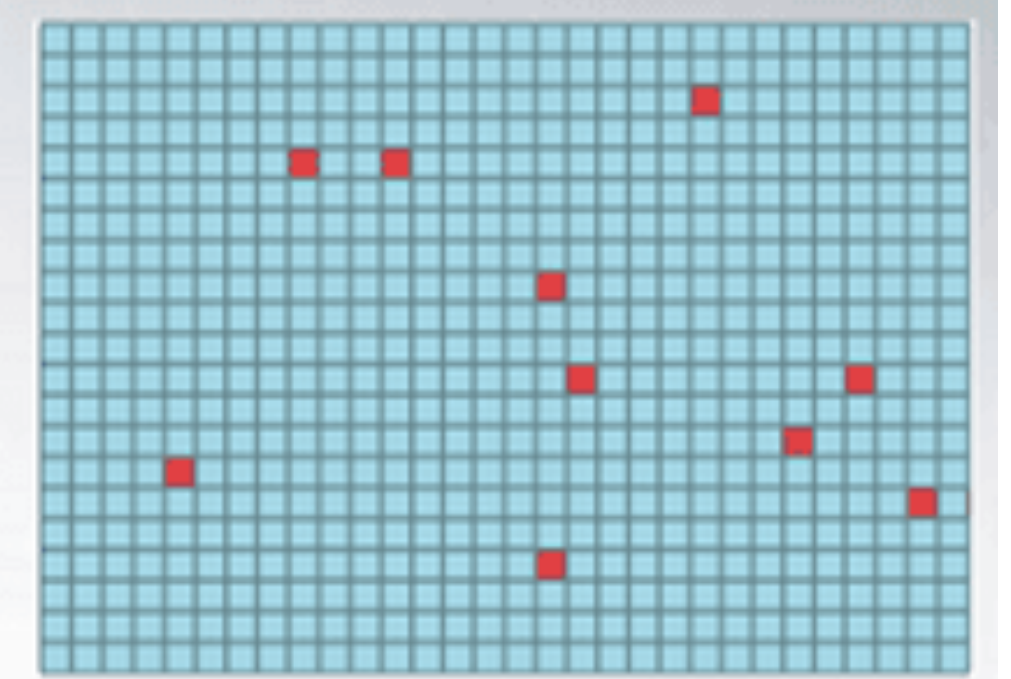
**Representing spatial data and
making maps like these**

Representing shapes on maps

- Vector data
 - Points, lines, polygons
- Raster data
 - Encodes the world as a continuous surface represented by a grid, such as the pixels of an image.



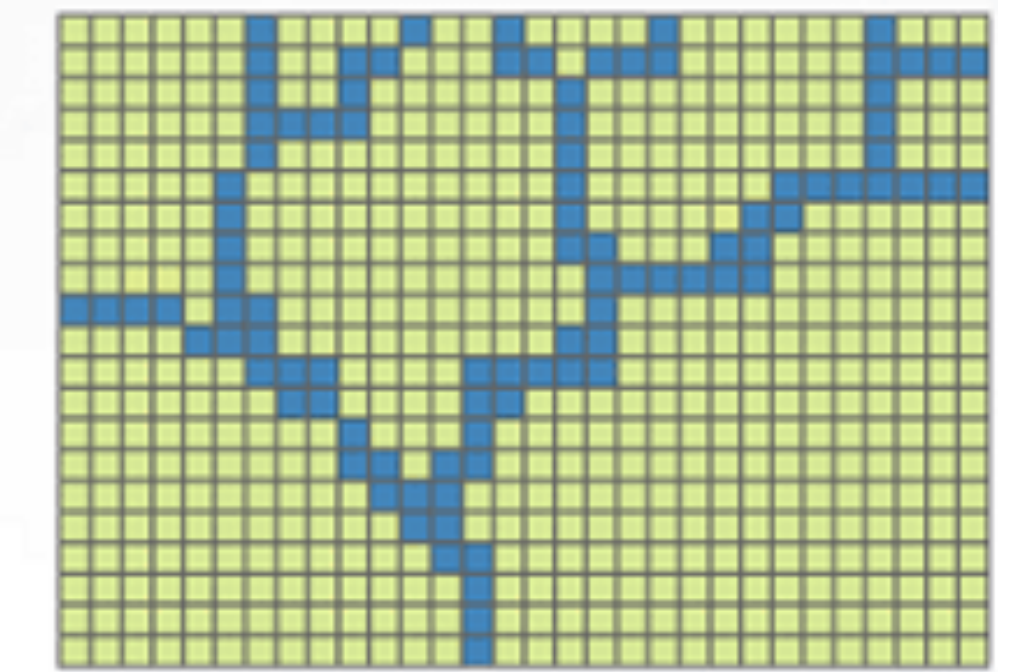
Point features



Raster point features



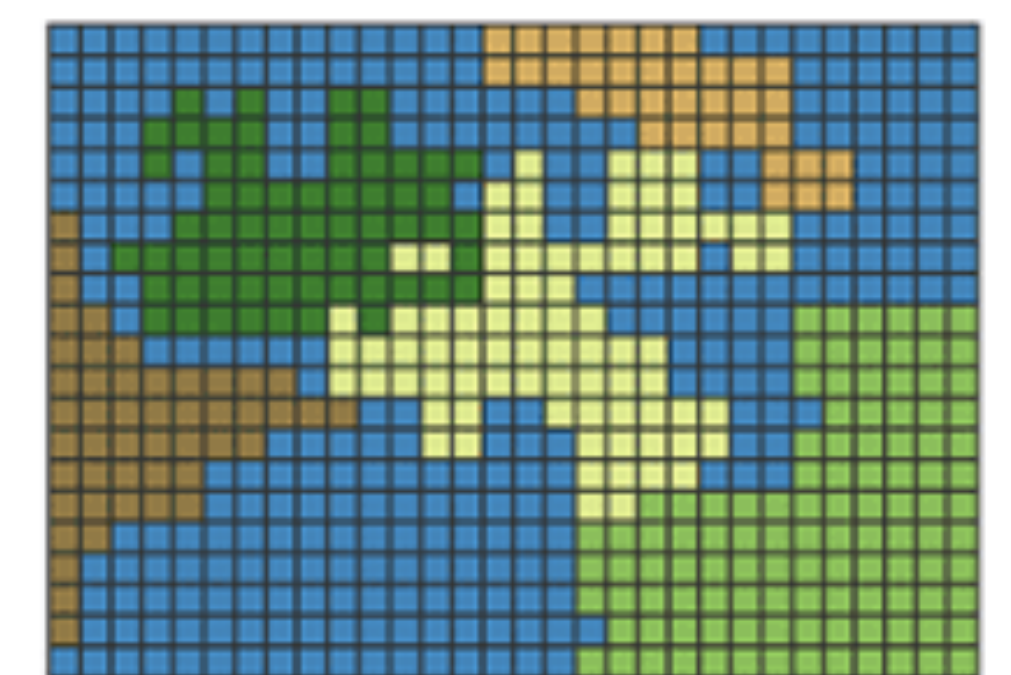
Line features



Raster line features



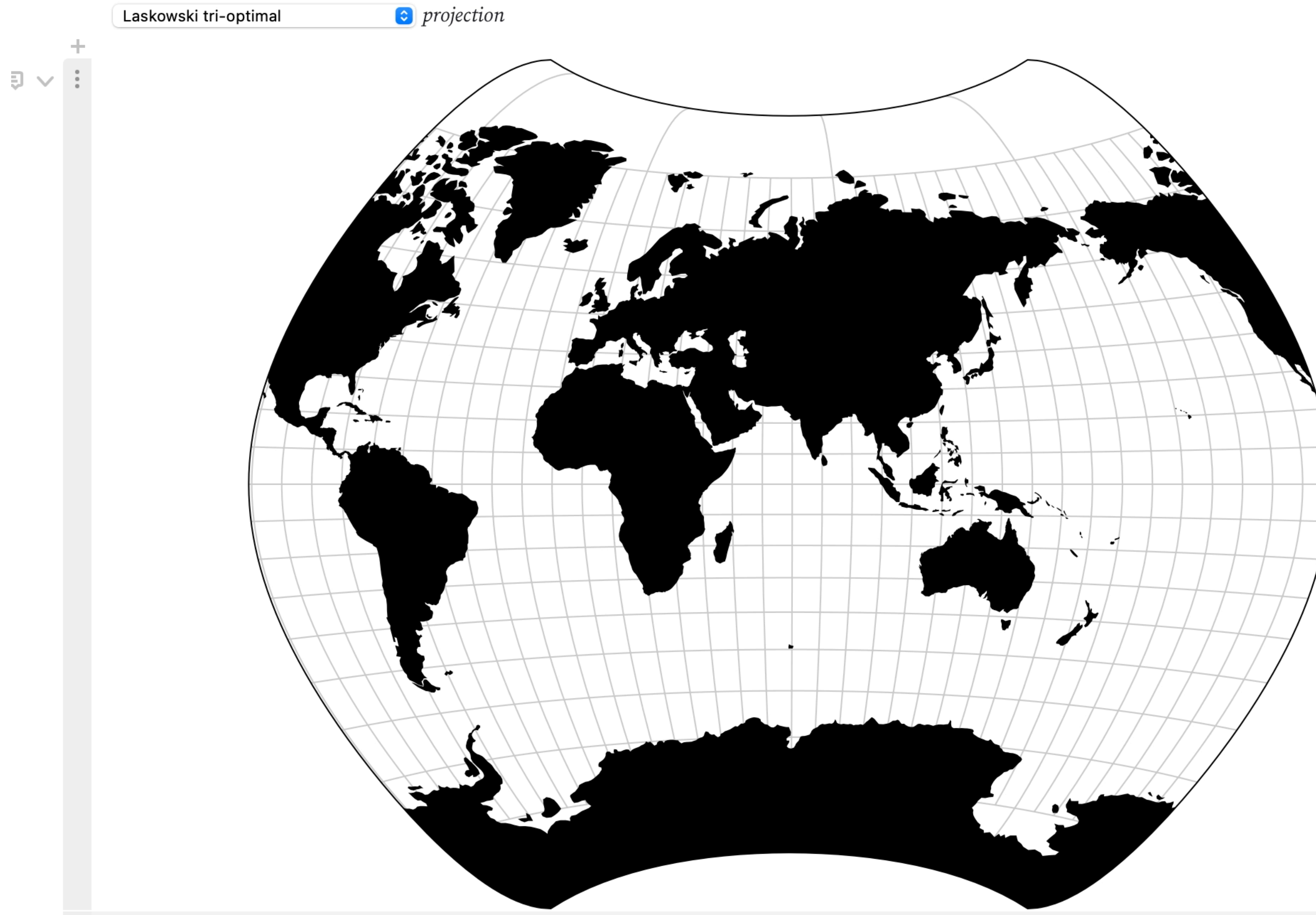
Polygon features



Raster polygon features

Projection Transitions

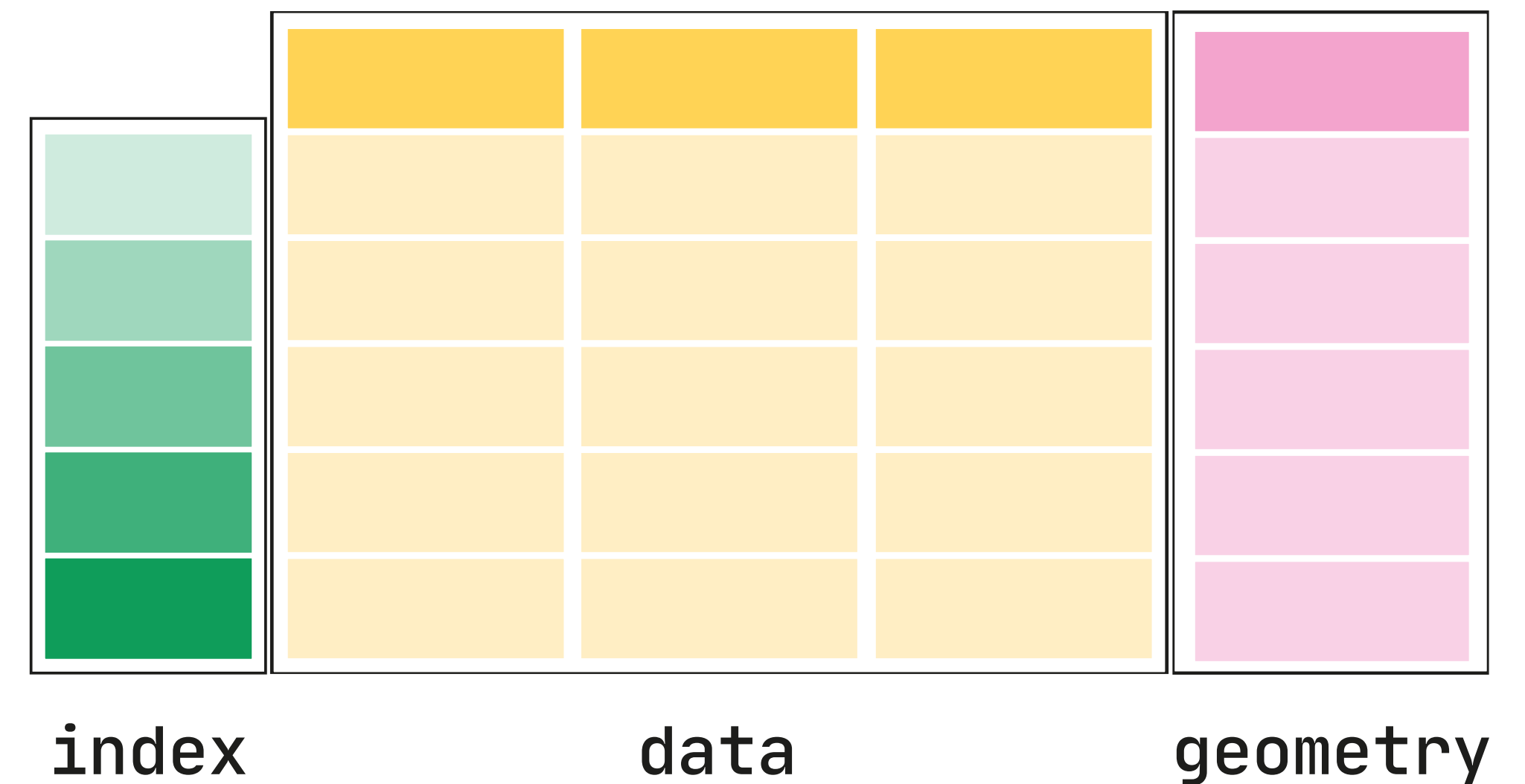
This notebook interpolates smoothly between projections; this is easiest when both projections are well-defined over the given viewport (here, the world).



Source: <https://observablehq.com/@d3/projection-transitions>

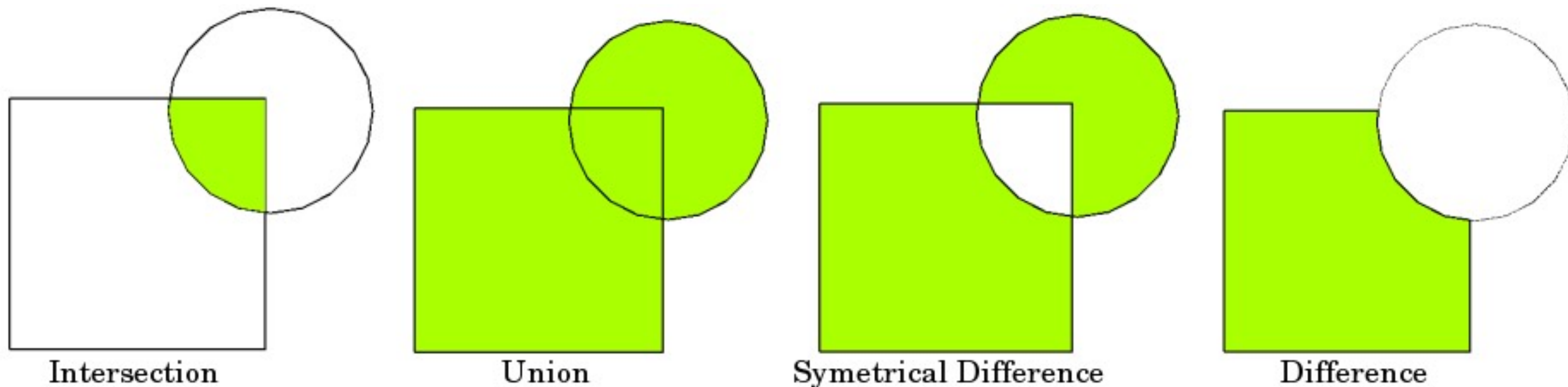
Geopandas

- Extends pandas, adds support for geospatial data
- Uses shapely library to represent vector geometry (e.g., POINT, LINE, POLYGON)
- Writes/reads shape files, GeoJSON and other formats
- Uses Coordinate Reference Systems (CRS) to represent how data aligns with the real world
- Different types of CRS: Geographic Coordinate Systems and Projected Coordinate Systems.



Geopandas skills

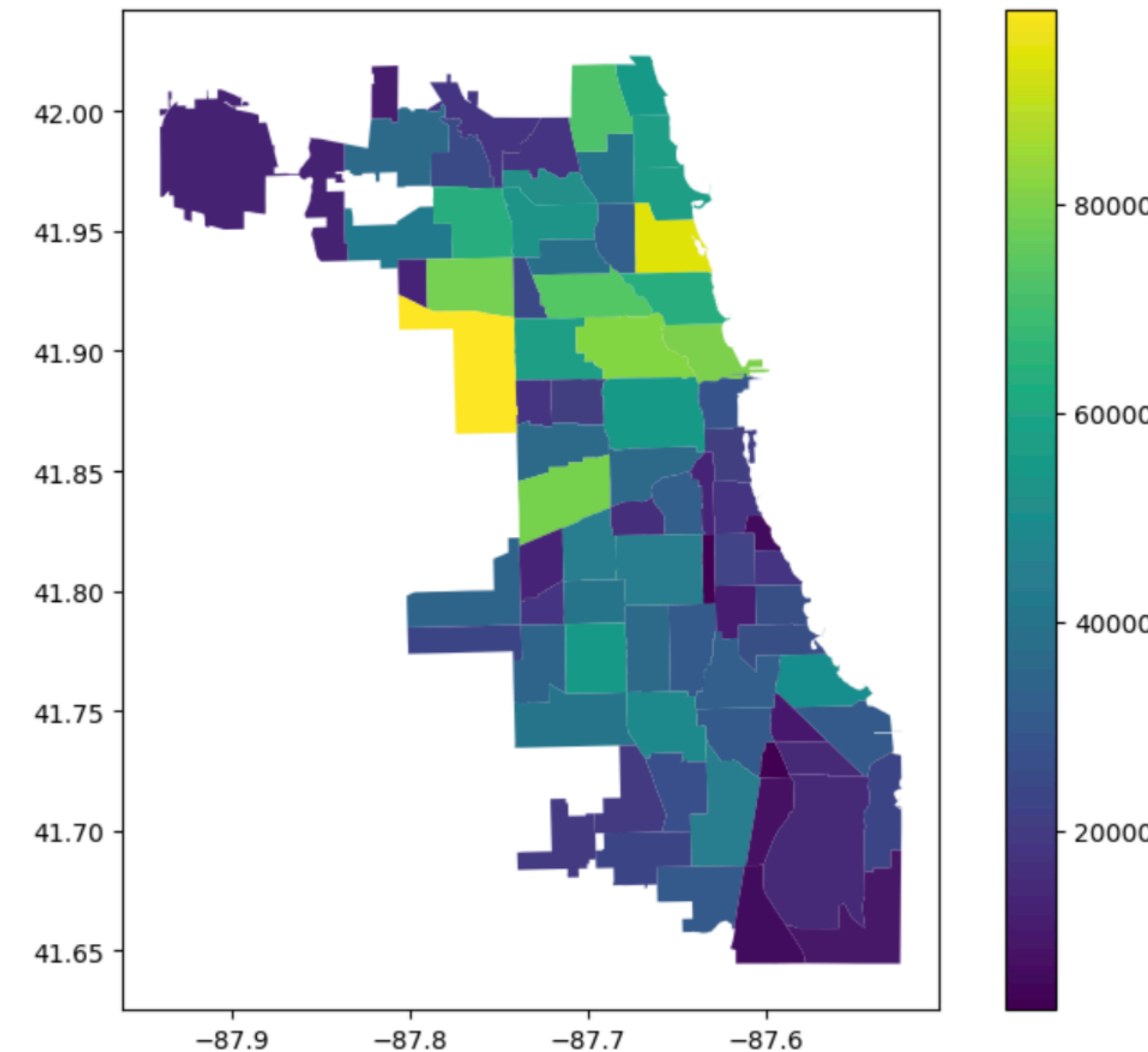
- Overlay spatial datasets: e.g. calculate what regions of Chicago are within 1km of a grocery store
- Join dataframes on shape information
- Aggregate across different subregions to get measurements at a more granular regional level



Geopandas tutorials

- https://geopandas.org/en/stable/getting_started/introduction.html
- https://geopandas.org/en/stable/docs/user_guide/mapping.html
- https://geopandas.org/en/stable/docs/user_guide/set_operations.html
- <https://geopandas.org/en/stable/gallery/index.html>

```
# Plot population estimates with an accurate legend  
In [7]: chicago.plot(column='POP2010', legend=True);
```



Visualizing Geospatial Data

You want to visualize how
many people have been
affected by COVID19
worldwide.

**Best approach to visualize
these data?**

[https://forms.gle/
8iSWt8MdJdfM7PC69](https://forms.gle/8iSWt8MdJdfM7PC69)



Project proposals

Due April 29

- Good research question that is unambiguous and detailed enough to tell us what data you will need and what kinds of analysis it takes to answer the question
- So think about how to get data, but also what you will do with it
- Brainstorm then refine

Random project ideas I will throw your way

- How many e-scooter accidents are there at UCSD? Are there more e-scooter accidents than e-bike?
 - EMS or hospital admissions data for UCSD Medical and / or Scripps Green (the two closest hospitals) e.g. <https://www.sandiegocounty.gov/content/sdc/ems/CoSD-LEMSIS.html>
- Is crime on campus changing year on year, in terms of both what crimes are committed and their rates? Are most crimes committed by other students or by outsiders? Are crime rates positively correlated with the student population size?
 - <http://ucsdpd.com> and its data is here <https://github.com/axelsagundo/ucsd-crime-logs-dataset>
- Is there a relationship between a student's sleep onset time, sleep duration, and their academic performance? Is there a relationship between their normal work time of day and their academic performance? (And/or also ask about wellbeing)
 - Setup a survey asking for GPA, typical work schedule for assignments, and upload of Apple Watch sleep data. If also wellbeing or depression there a variety of short surveys you can include

Random project ideas an AI will throw your way

Qwen 3.5 running in Ollama on my laptop (YMMV)

- Prompt: “You are a professor teaching a datascience introductory course. You are generating ten example ideas for datascience projects. Each project should include the following aspects: 1) an unambiguously worded question defining exactly the scope of the project and defining the metrics and analysis that will be performed; 2) a hypothesis about which way the results will turn out; 3) the project idea should be based on publicly available data and should include a source where this data may be obtained.”
- You could probably do better than this prompt and certainly can do better than such an AI small enough to run on a laptop.
- These responses are all
 - too small in scope of the project and the question wording reads more like a recipe than a datascience question. In many cases they could be refined into something good!
 - All based on very generic projects that have been done to death on Kaggle. Compare with the ideas on the last page that are local and unique/personal

Random project ideas an AI will throw your way

Not worded as a question, idea OK, scope too small and this version of Spotify data too limited

- Question: "Using a dataset of top Spotify tracks, categorize songs by genre and calculate the average track duration for each genre. Then, compare the duration of songs with higher listen counts against songs with lower listen counts to identify if there is a 'sweet spot' duration for popularity."
- Metrics: Group means (duration by genre), Mean vs. Median comparison.
- Hypothesis: Shorter tracks (approx. 3-4 minutes) will have a higher average listen count than tracks significantly shorter or longer.
- Data Source: Kaggle (Spotify Top 50 Global Tracks).

Random project ideas an AI will throw your way

Not worded as a question, idea OK, scope too limited but dataset is huge

- Using New York City Taxi and Limousine Commission (TLC) trip data, filter for trips taken during rainy days versus sunny days. Compare the average trip duration and total trip count for each condition to quantify the impact of weather on ride demand and traffic flow.
 - Metrics: T-test (independent samples), Mean trip duration by weather type.
 - Hypothesis: Rainy days will show a 15% increase in average trip duration due to slower driving speeds, but a decrease in total trip count due to lower demand for non-essential travel.
 - Data Source: NYC Open Data Portal (Taxi and Limousine Commission).